



FABRIC STRUCTURE & DESIGN

Part: B

Course Code: FE 305-0723

Fancy Designs

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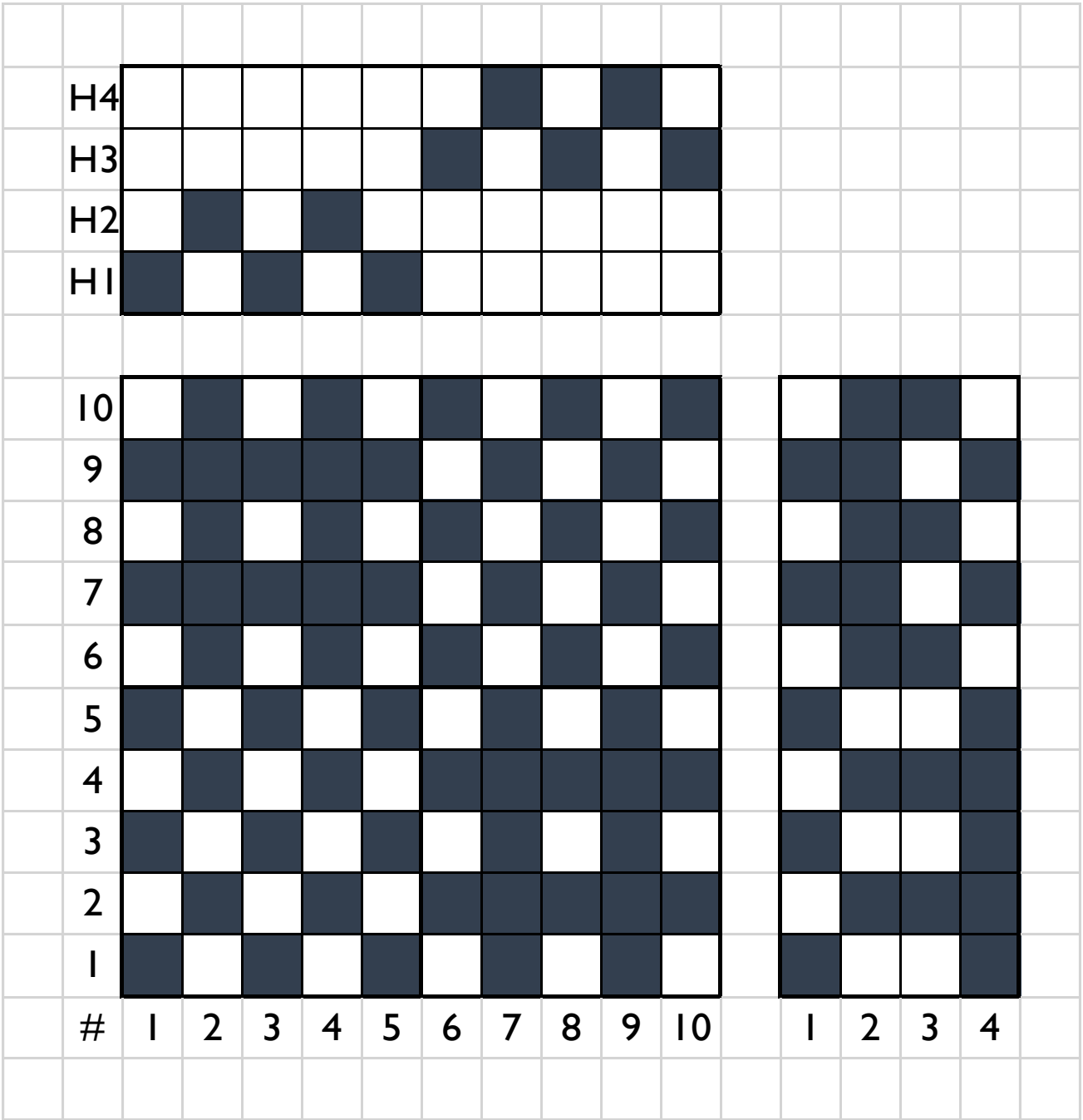
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HUCKABACK WEAVE

Features

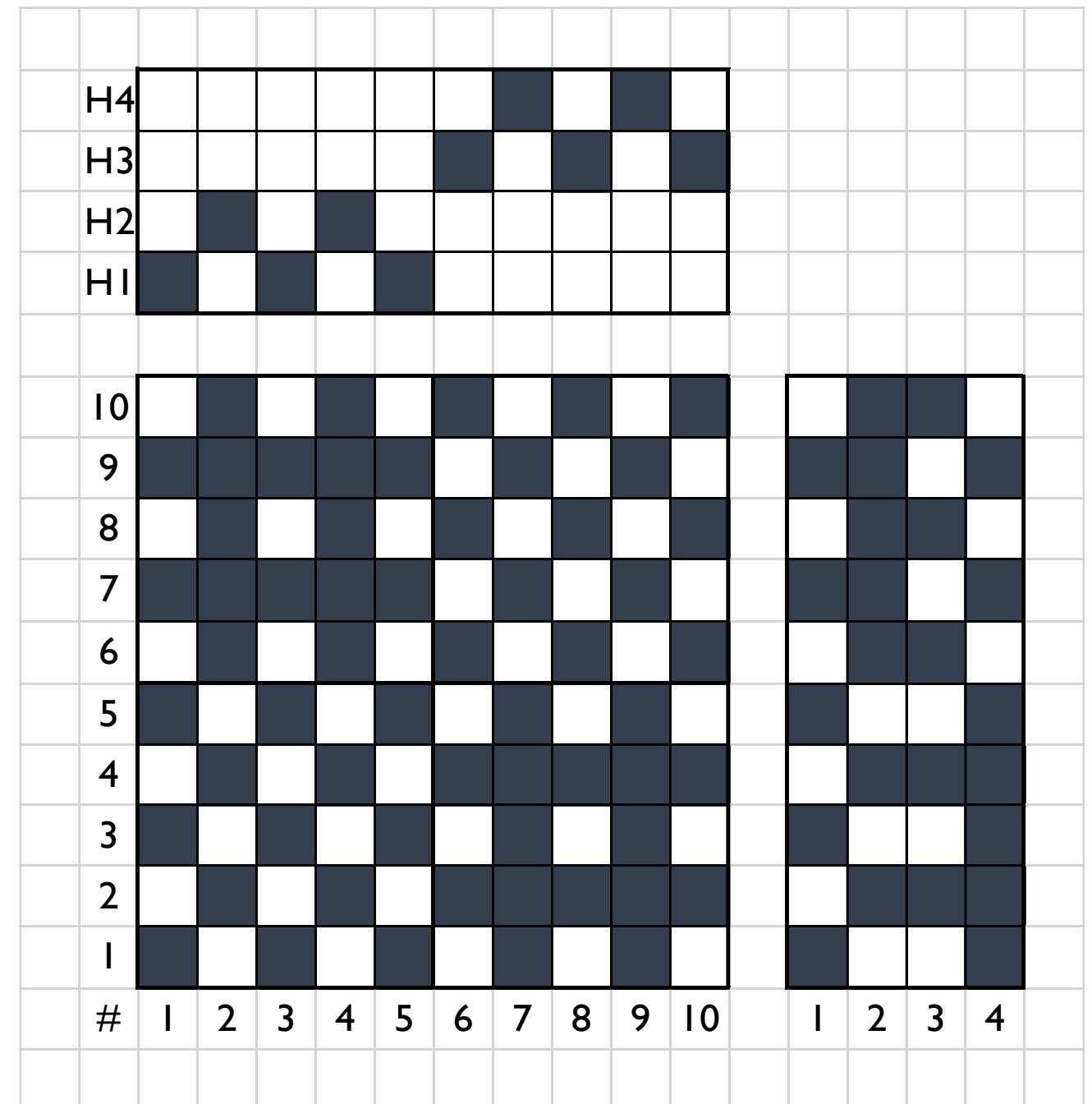
- 1. Generally applied in the manufacturing of *non-pile towels*.
- 2. Rough surface which is produced by floating threads in group.
- 3. Balanced huckaback is produced if repeat size (RS) is twice an odd number. (i.e. $2 \times 5 = 10$; RS = 10 Ends x 10 Picks).
- 4. However, the weave can be made with a repeat based on twice an even number of thread. (UB)
- 5. 10 x 10 is the widely used repeat size.
- 6. This weave can be divided diagonally into equal parts.



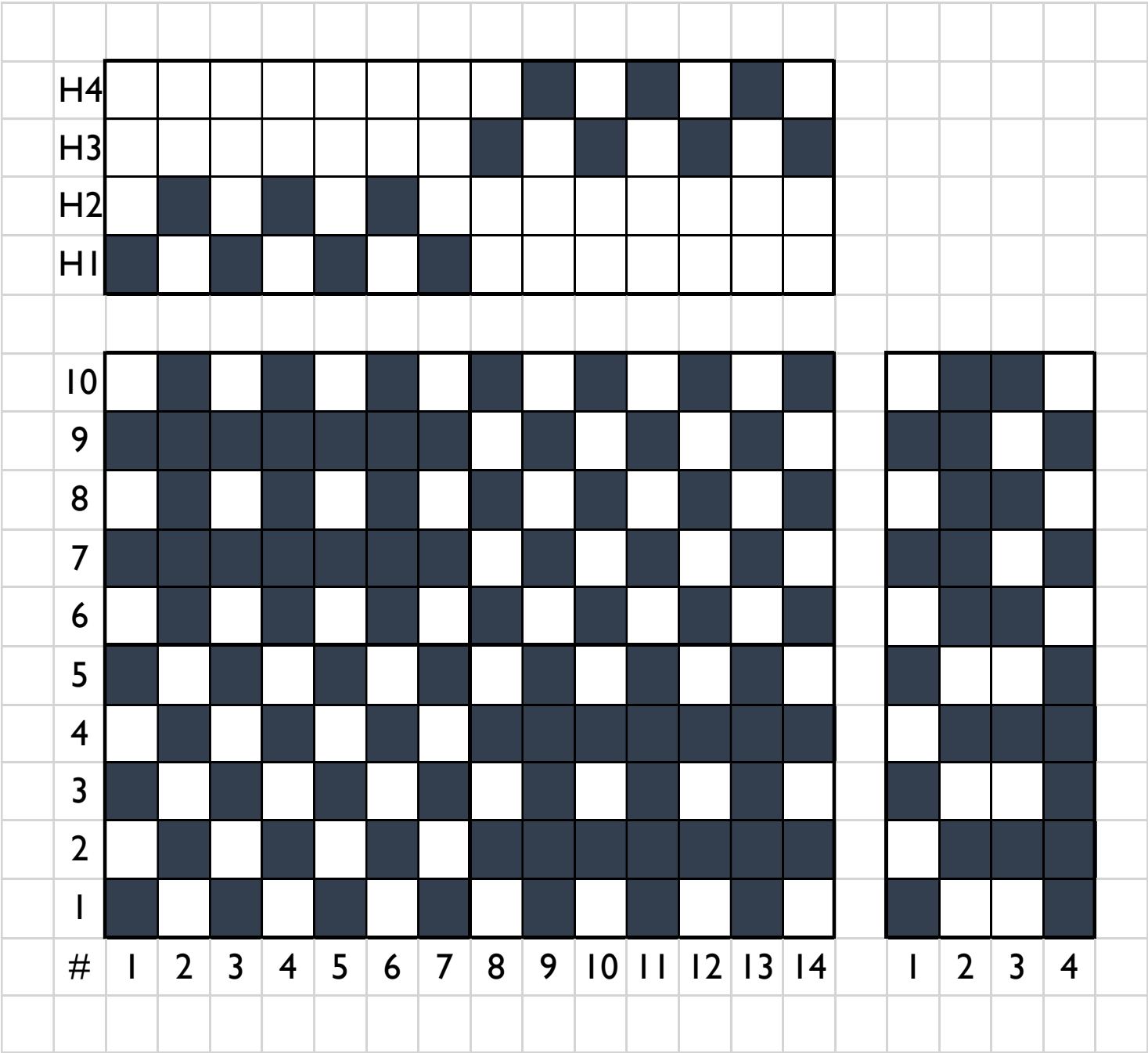
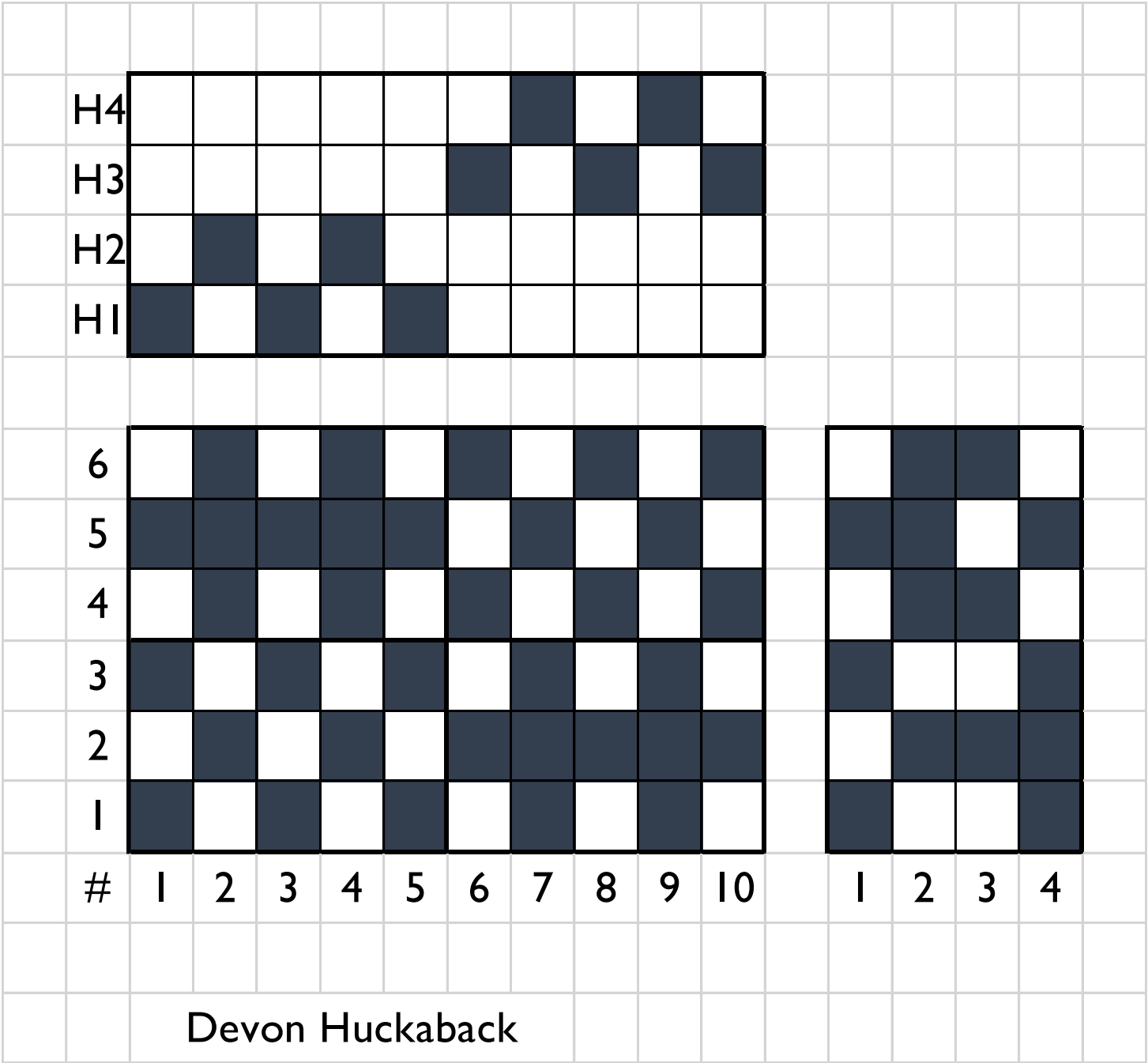
CONSTRUCTION PRINCIPLES

Stages in constructing the Huckaback weave:

1. Mark out the repeat size, divide into quarters and fill in plain weave in two opposite ones as shown at first.
2. Fill in a motif in the other two quarters, which is preferably produced by taking plain weave and adding or removing some lifts, as at second one; care should be taken to ensure that the motif and the plain weave bind together effectively.
3. The final weave is produced by combining first and second one.



HUCKABACK WEAVE



HUCKABACK WEAVE

Uses

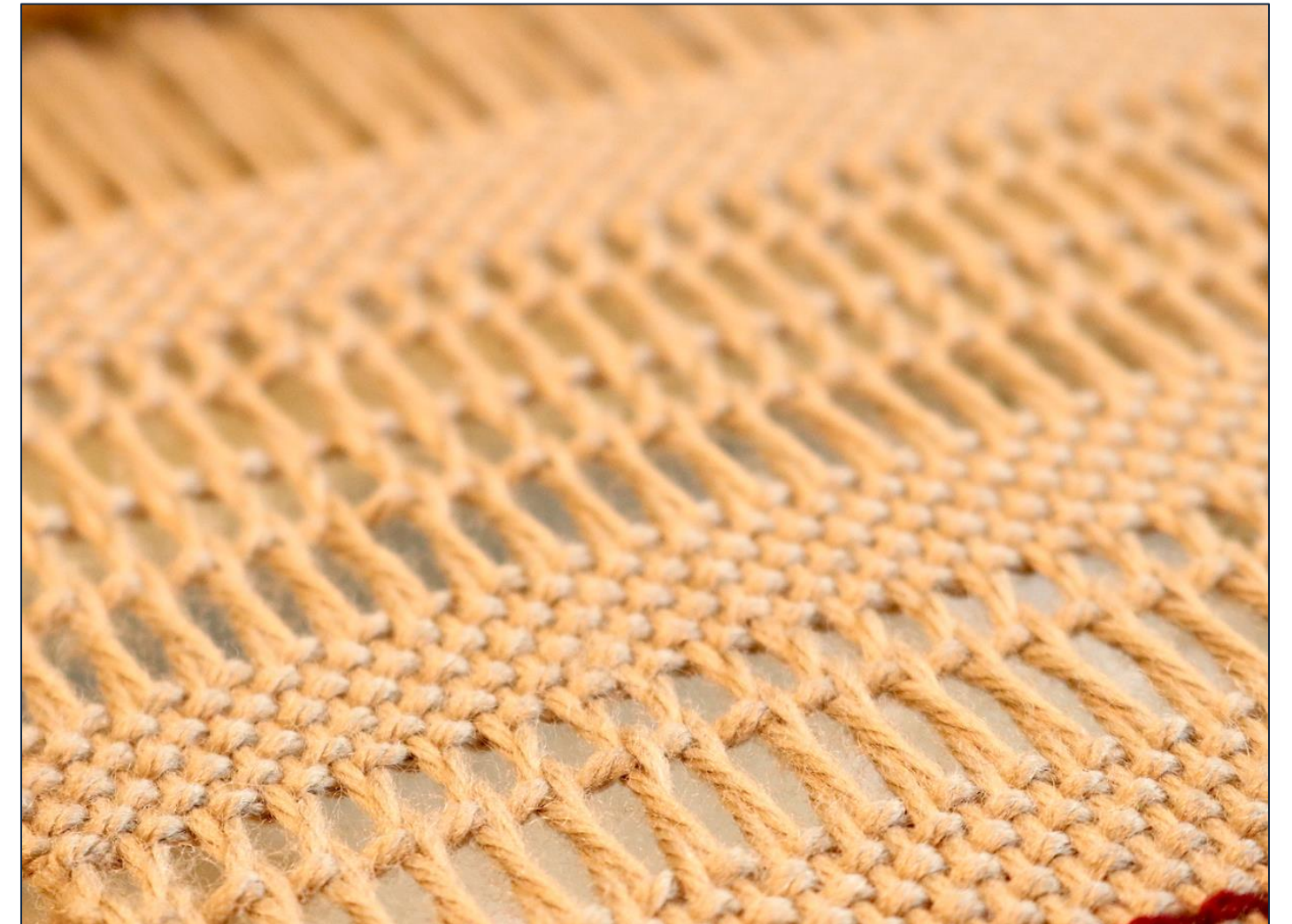
- Cotton, Linen yarns are commonly used.
- Coarser qualities: Suitable for Hand towel, Glass cloths, Roller towels.
- Finer qualities: Suitable for Shirts, Dress-wear, Table Linen.



MOCKLENO WEAVE

Features

1. An *open perforated weave*, similar to *leno fabrics*.
2. It looks similar to Huckaback, but the *denting method* is different.
3. Some threads work in plain weave, while other *threads float on the face or back of the fabric*.
4. The threads in each group are, whenever possible, *placed in the same dent*. This brings the floating threads close together and creates *small gaps or openings* in the fabric.
5. These openings make the fabric *look like a gauze* or leno fabric, so it is called “*mock leno*.”
6. An even number of threads is normally used in one repeat.

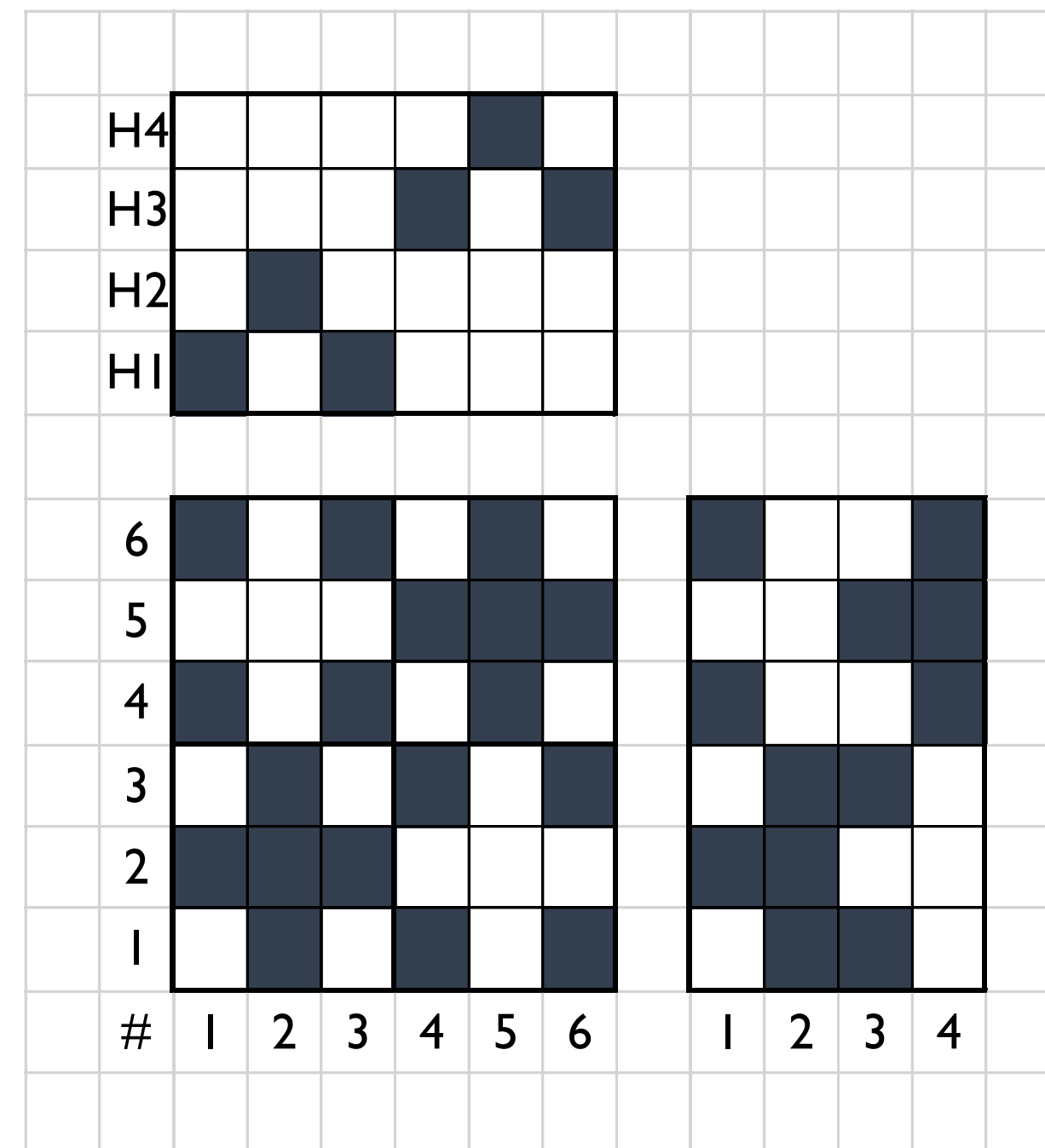


MOCKLENO WEAVE

Features

7. The weave can also be *divided diagonally into two equal parts*.

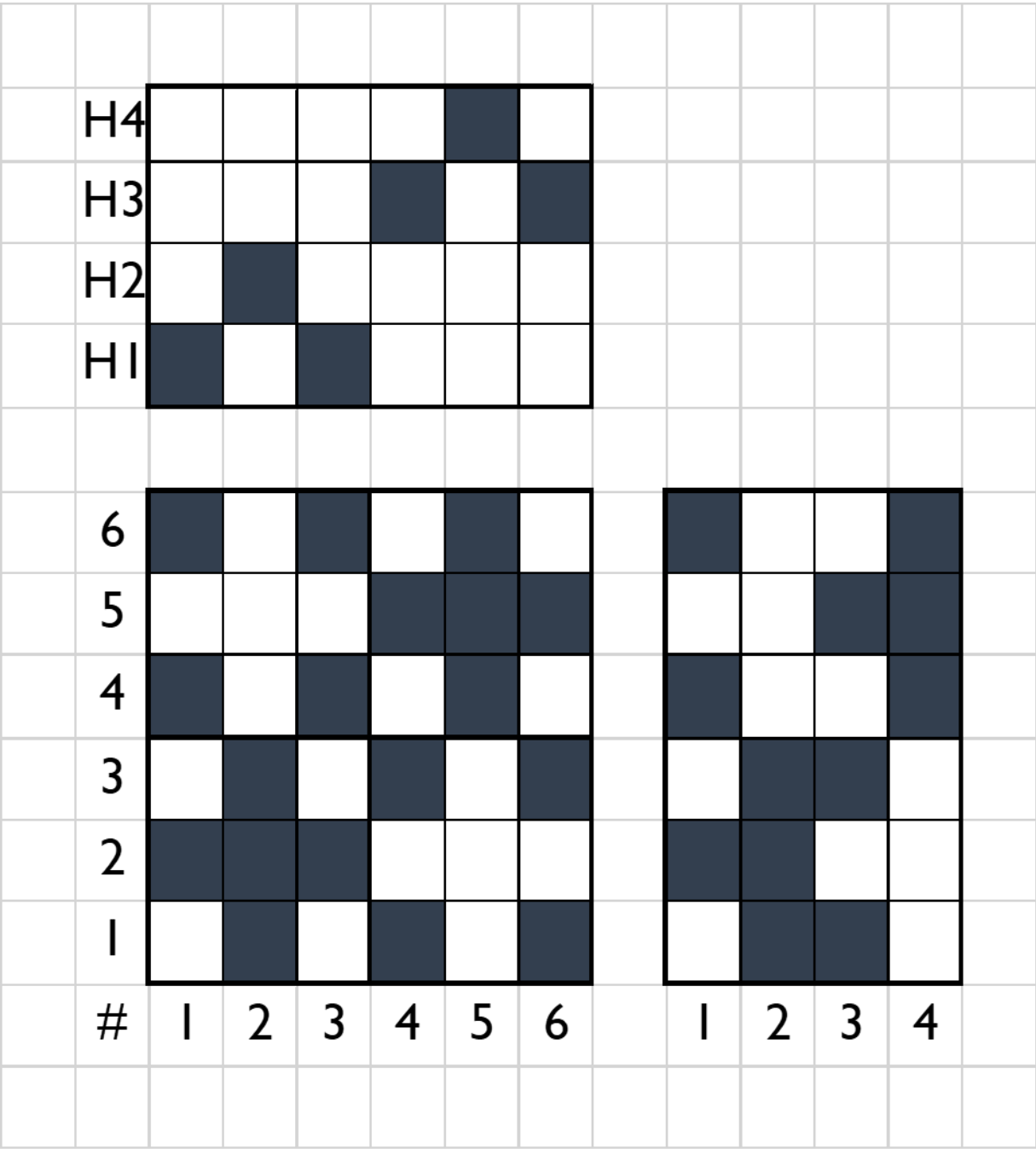
8. The smallest repeat size is 6 x 6.



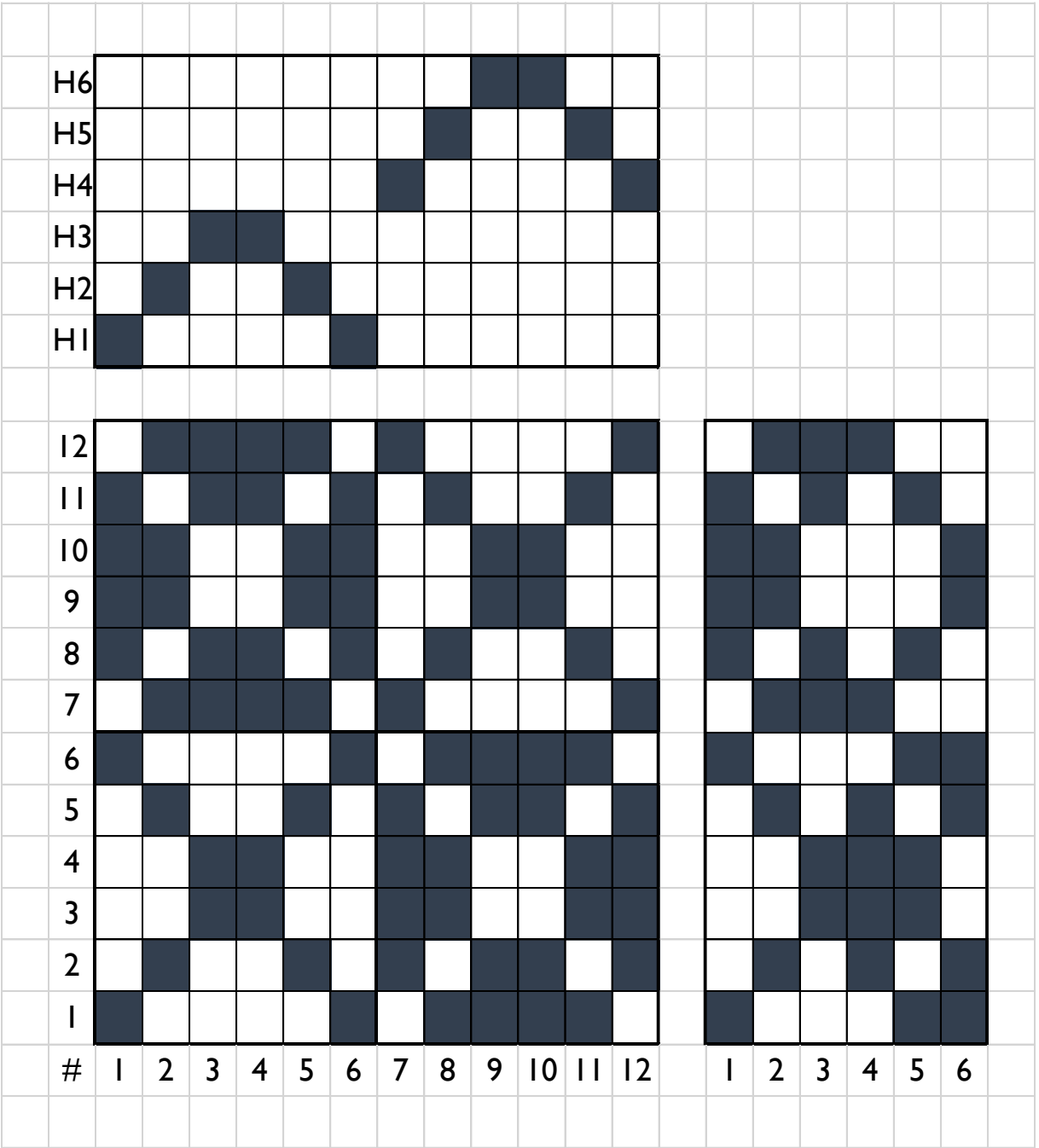
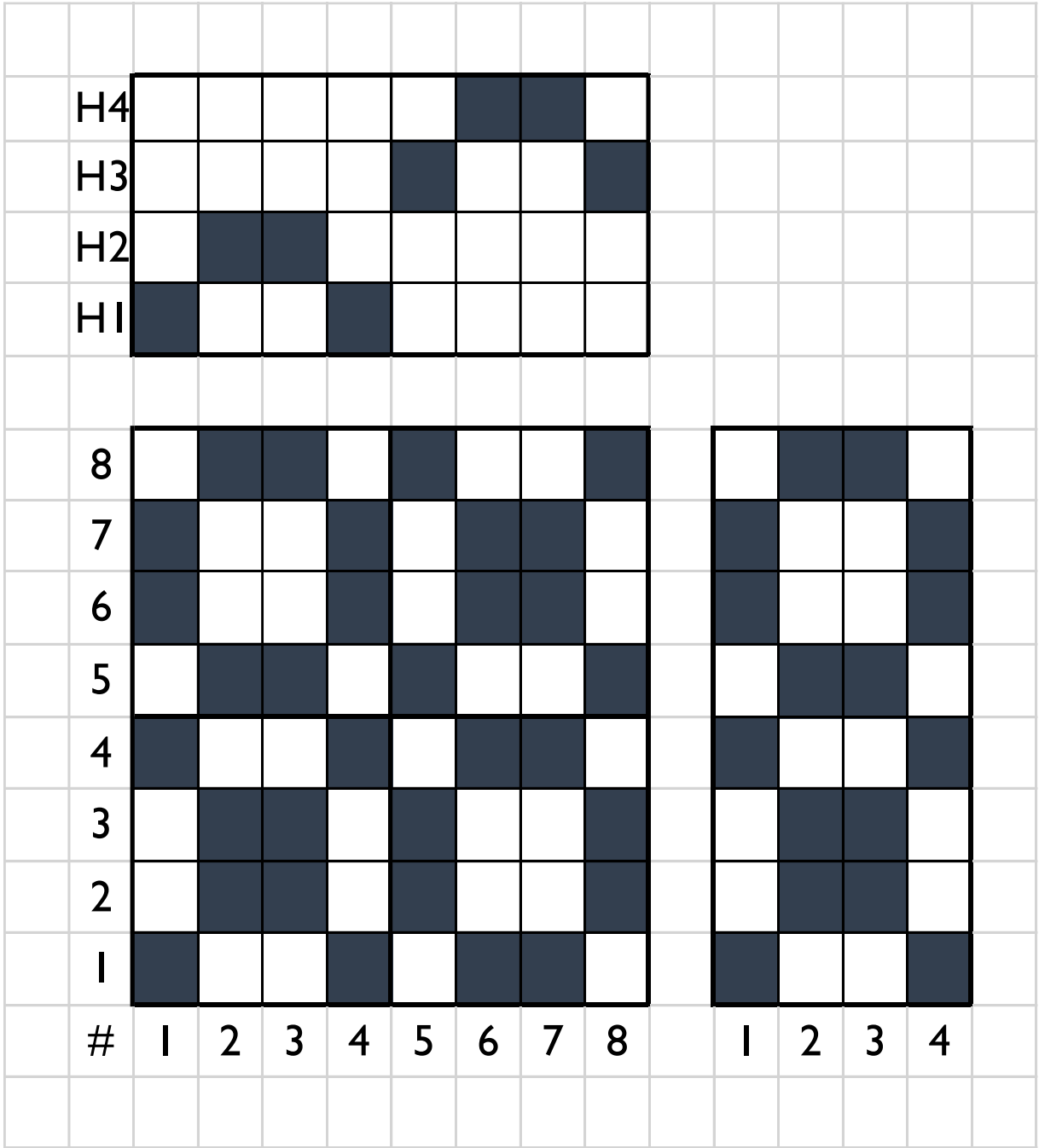
CONSTRUCTION PRINCIPLES

Stages in constructing the Mockleno weave:

- 1. Mark out the repeat size, divide into quarters and fill a small motif in opposite quarters, as in first step.
- 2. Completely reverse this motif in the two remaining quarters, by substituting warp lifts for weft lifts and vice versa, as in second step.
- 3. Combine first and second steps to give the final weave.



MOCKLENO WEAVE



MOCKLENO WEAVE

Uses

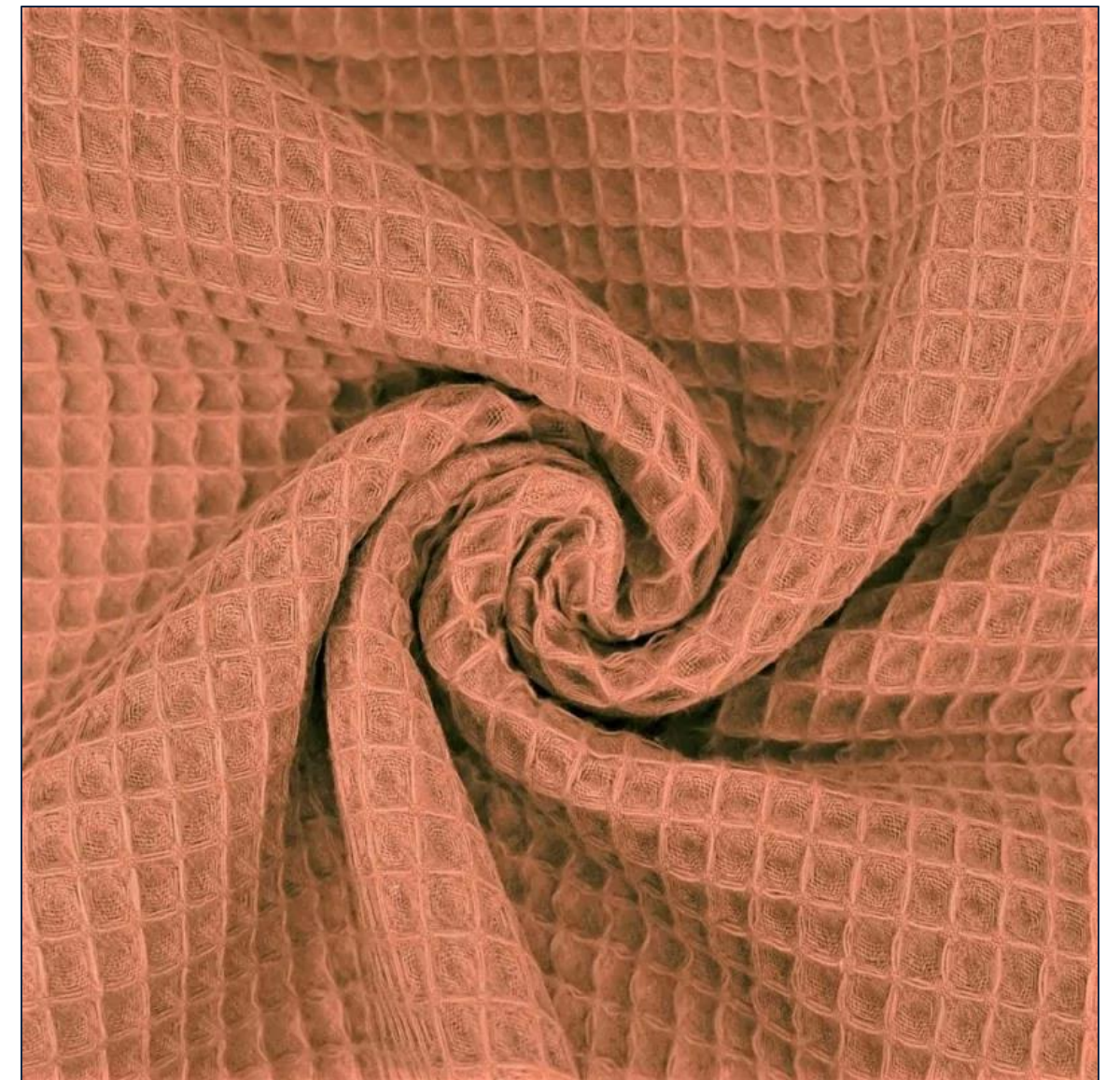
- Curtain, Table Linen, Canvas Cloth.
- Apparel Fabric, Embroidery Cloth, Brocades, Blouses, Dress-wear.



ORDINARY HONEYCOMB

Features

1. Alternate *raised and sunk diamond shaped* areas which give the effect of honeycomb.
2. *Both side* of fabric looks *same*.
3. *Surface* of the fabric is *rough*.
4. Long floats of warp and weft yarns.
5. In repeat size number of ends and picks may be equal or unequal and multiple of two.
6. In a larger repeat, a *double row of binding* is used. $1\ 1 / 1\ \alpha$ twill is used first. This makes the fabric stronger and more stable.
7. *Pointed drafting* system is normally used.



CONSTRUCTION PRINCIPLES

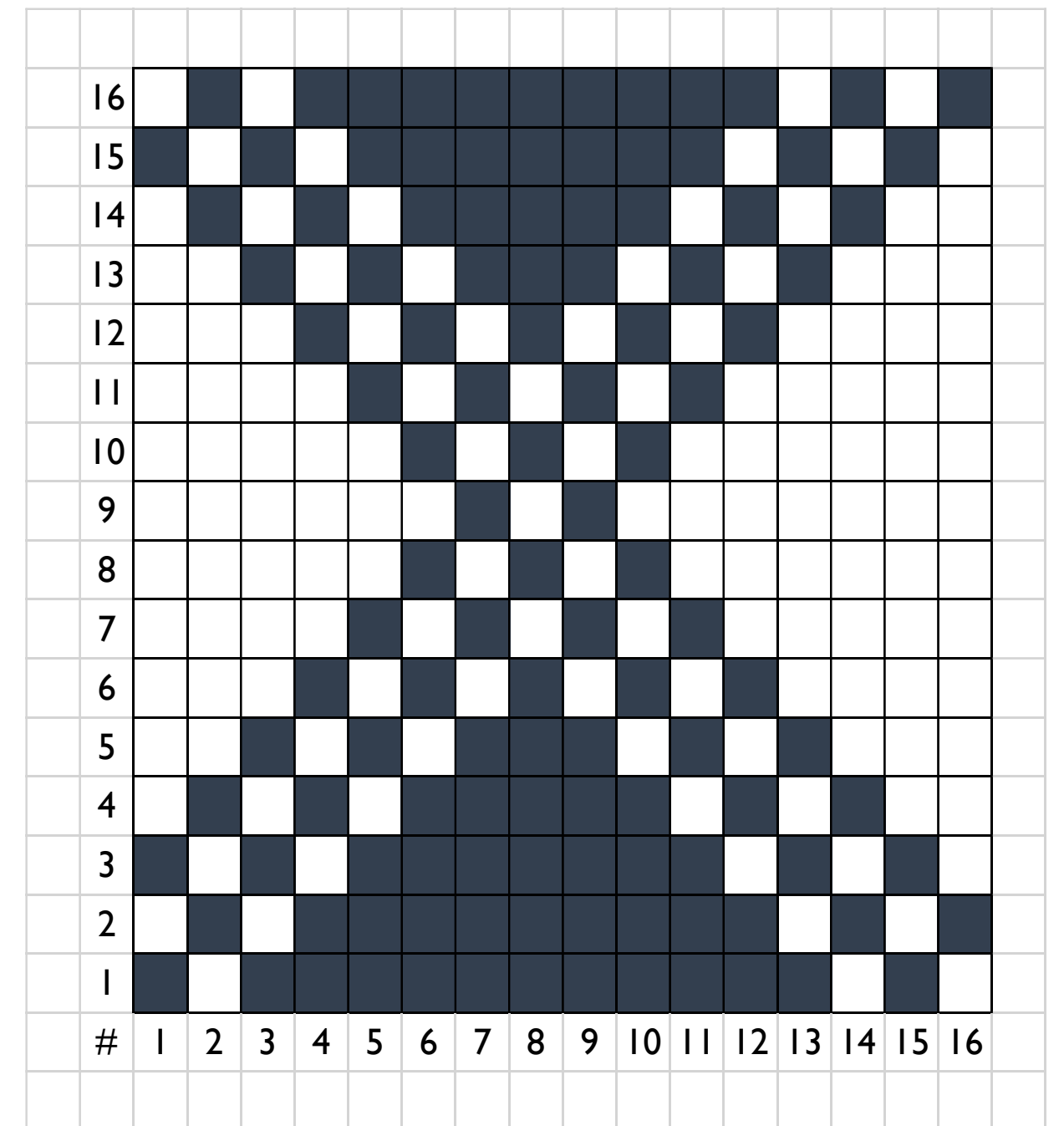
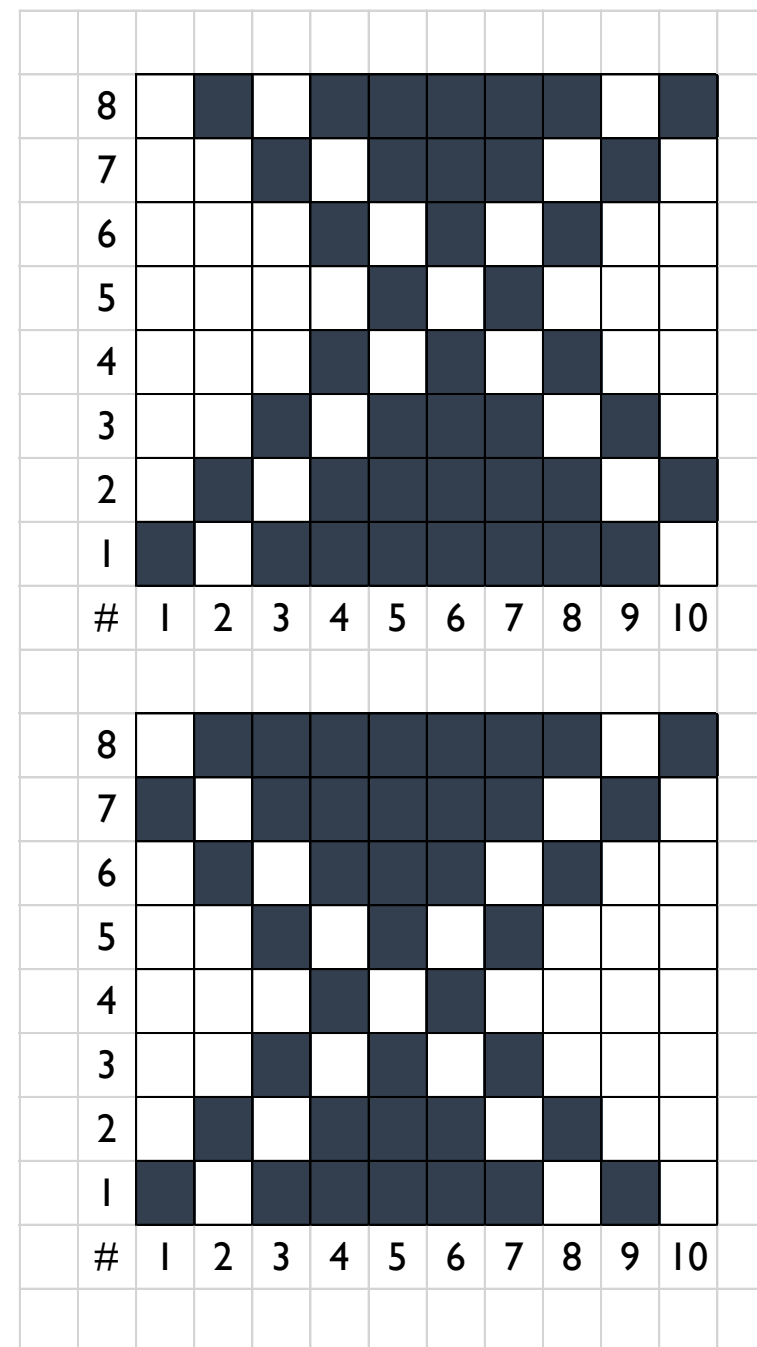
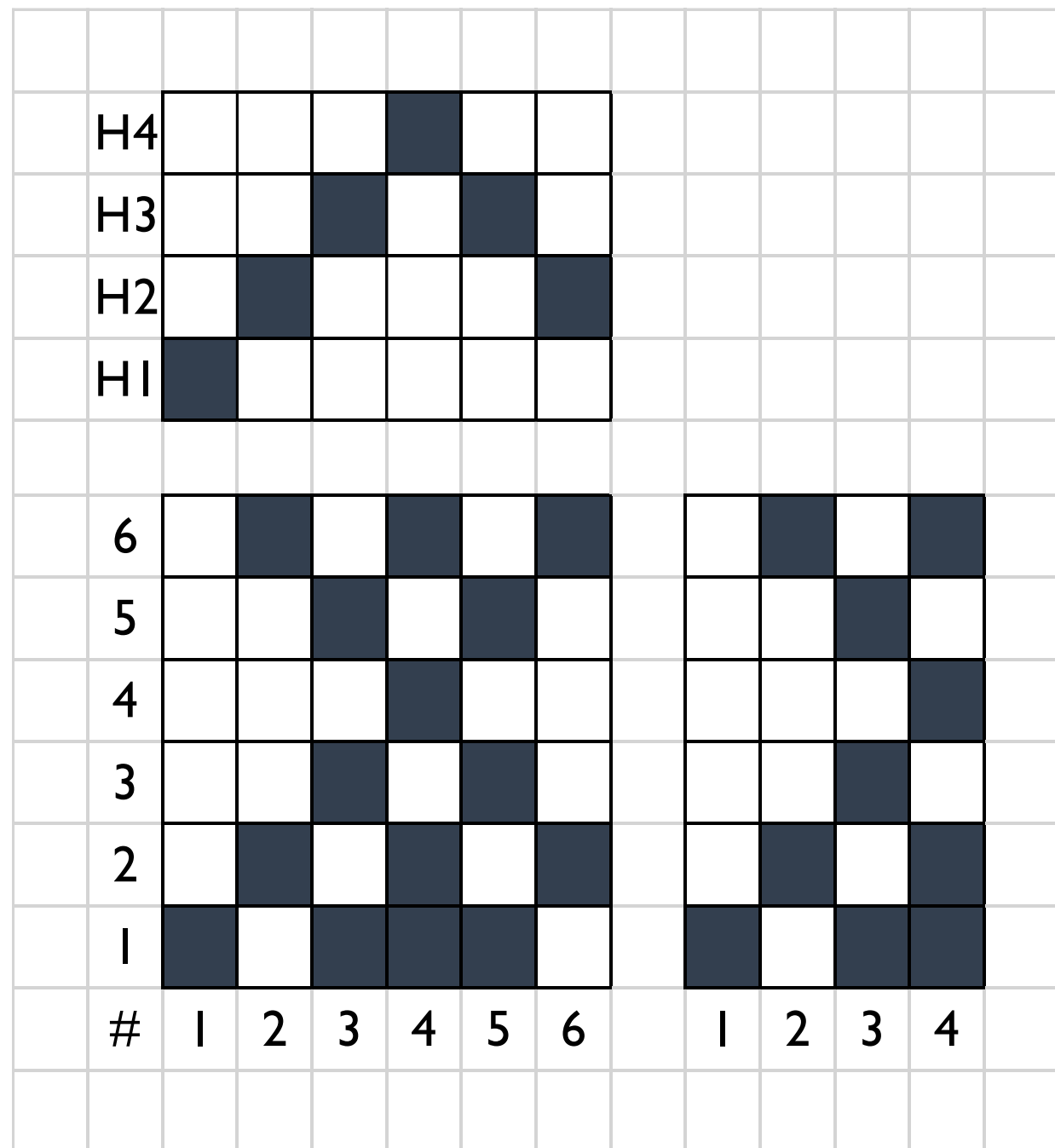
Stages in constructing the Ordinary Honeycomb weave:

- 1. Construct a 1/α Z twill starting in the bottom left-hand corner, then a similar one running in the opposite direction and starting one square inf or one square down from the top left-hand corner, so that there will be a clean intersection of the twill lines, as at first stage.
- 2. In one of the two diamonds produced, leave a row of stitching points and then lift the remainder of the diamond solid. This is the final weave.

	6							
	5							
	4							
	3							
	2							
	1							
	#	1	2	3	4	5	6	

	6							
	5							
	4							
	3							
	2							
	1							
	#	1	2	3	4	5	6	

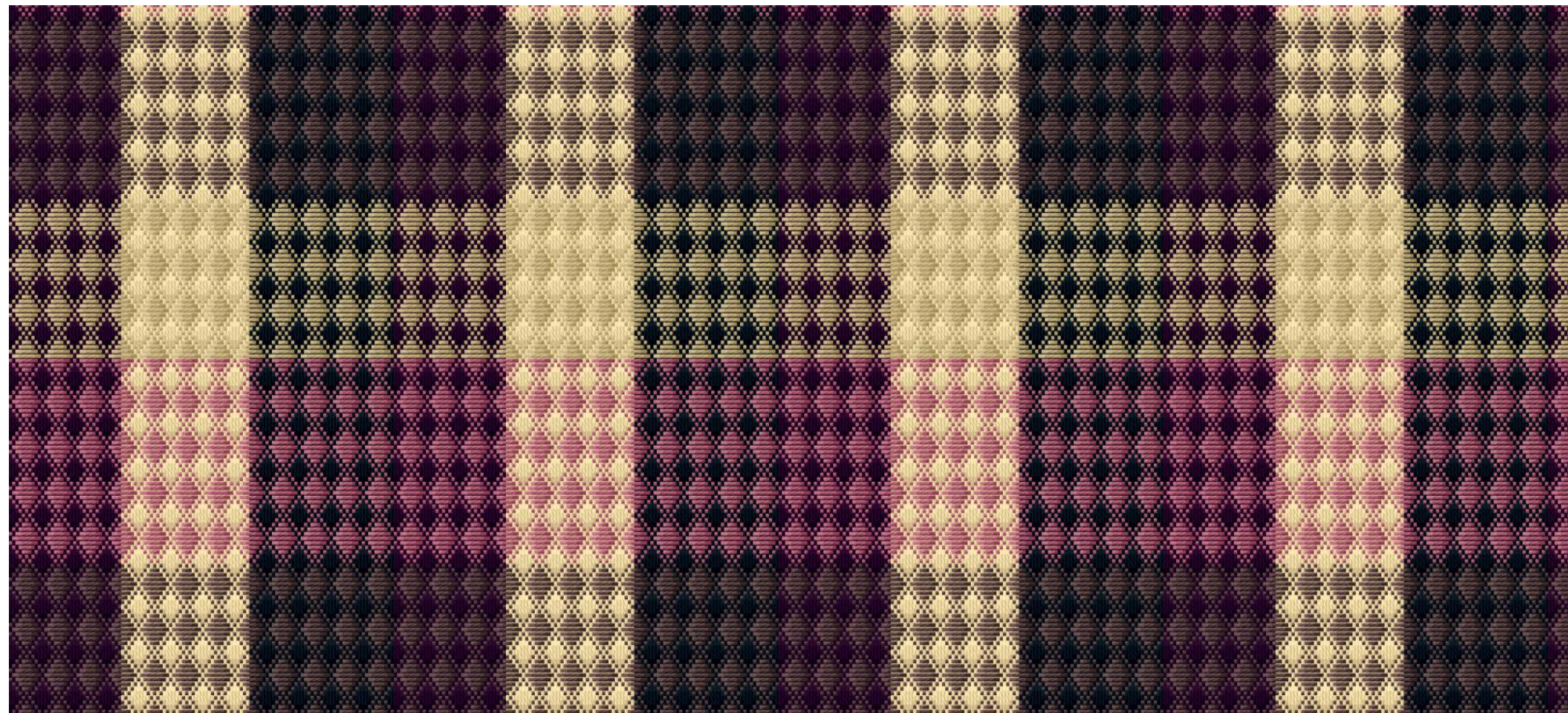
ORDINARY HONEYCOMB



ORDINARY HONEYCOMB

Uses

- Towel, Glass cloths, Bath mats where moisture absorption properties are desired.
- Coarser qualities: Quilts, Blankets.
- Finer qualities: Suitable for Shirts, Brocades.



BRIGHTON HONEYCOMB

Features

1. Complex structure.
2. Varying size of *more honeycomb cells in this weave*.
3. Surface of the fabric is rough like ordinary honeycomb.
4. No of threads per repeat should always be a *multiple of four*. (i.e. 12 ends x 12 picks).
5. Longest float should always be *one less than half the number of thread in repeat*. (i.e. $12/2 - 1 = 5$)
6. *Both side of fabric looks same* like ordinary honeycomb.
7. *Straight drafting system* is used.

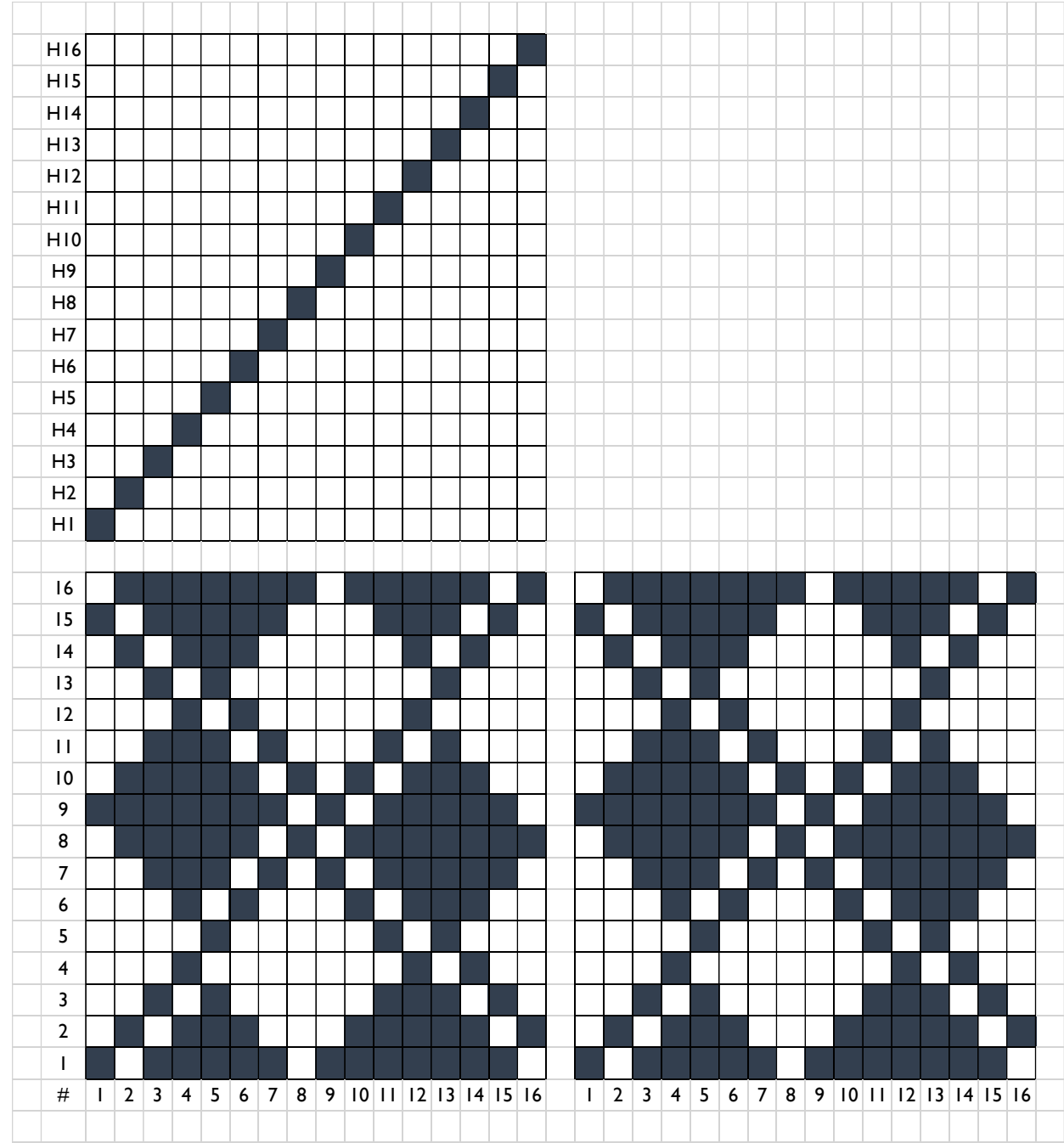
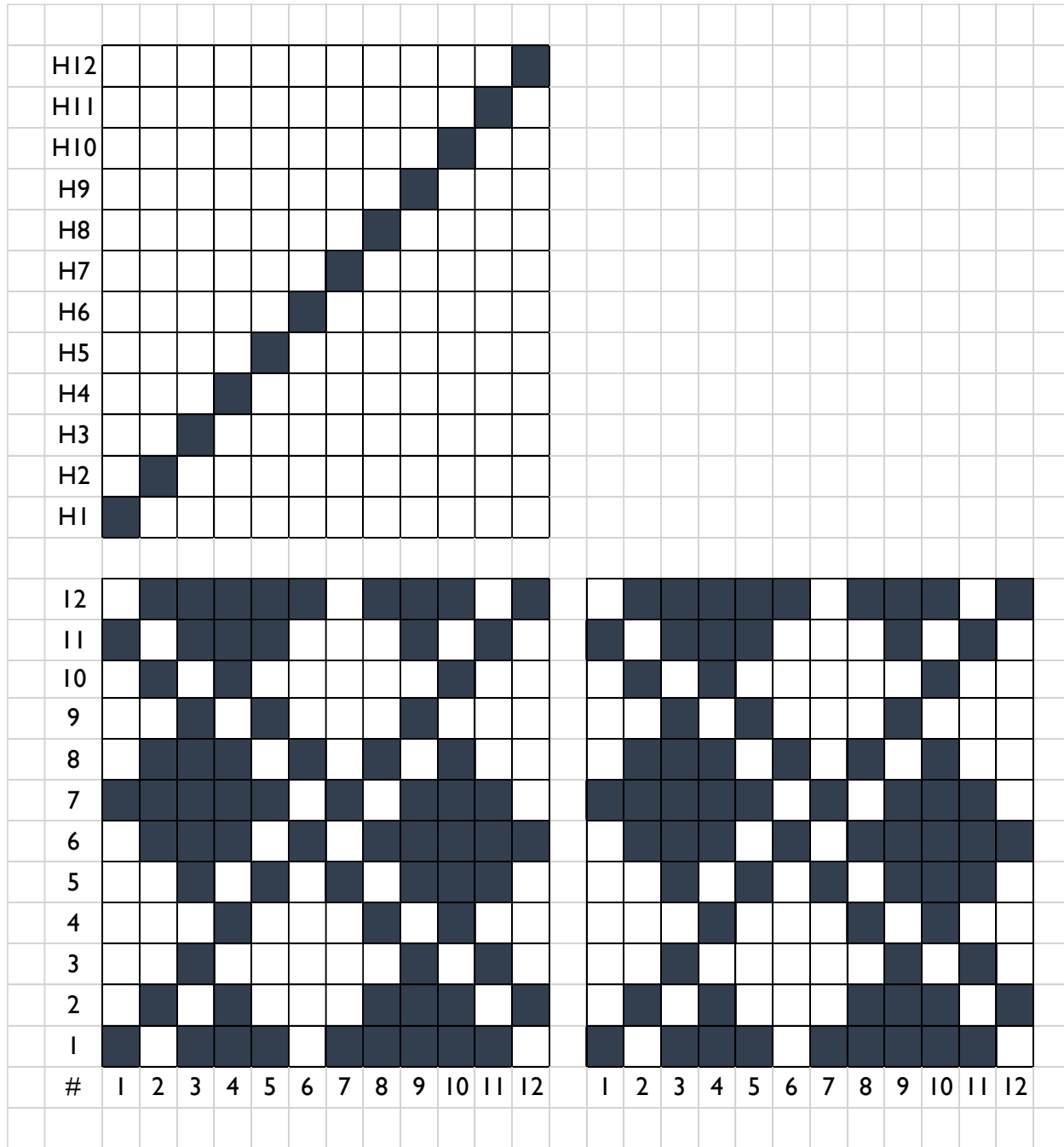
CONSTRUCTION PRINCIPLES

Stages in constructing the Brighton Honeycomb weave:

1. Construct a $1/\alpha$, Z twill, starting in the bottom left-hand corner, and then construct $1\ 1 / 1\ \alpha$, S twill starting with the first warp lifts in the squares to the right and below the square in the top left-hand corner, and indicate the points on the double row of binding which are immediately adjacent to those of intersection that will allow extensive floats in the weft direction, as illustrated in first stage.
2. Using the points indicated in first stage as the extreme lift of the longest float, lift the remaining adjacent ends, as in second stage.
3. Each of these warp floats now form the center float of a diamond which can be completed. This is the final weave.

12												
11												
10												
9												
8												
7												
6												
5												
4												
3												
2												
1												
#	1	2	3	4	5	6	7	8	9	10	11	12

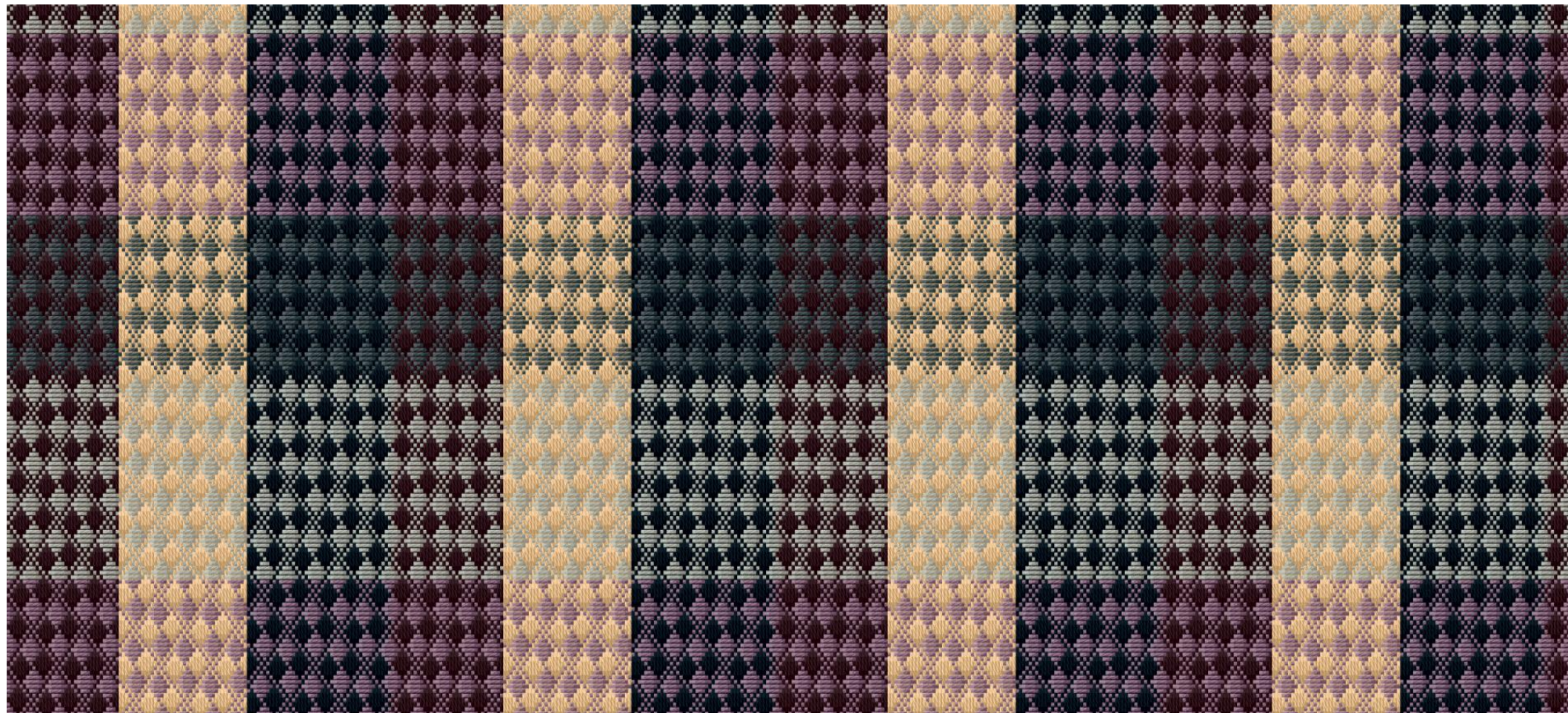
BRIGHTON HONEYCOMB



BRIGHTON HONEYCOMB

Uses

- Not popular as Ordinary Honeycomb.
- Hand towel, Crockery towels, Glass cloths.



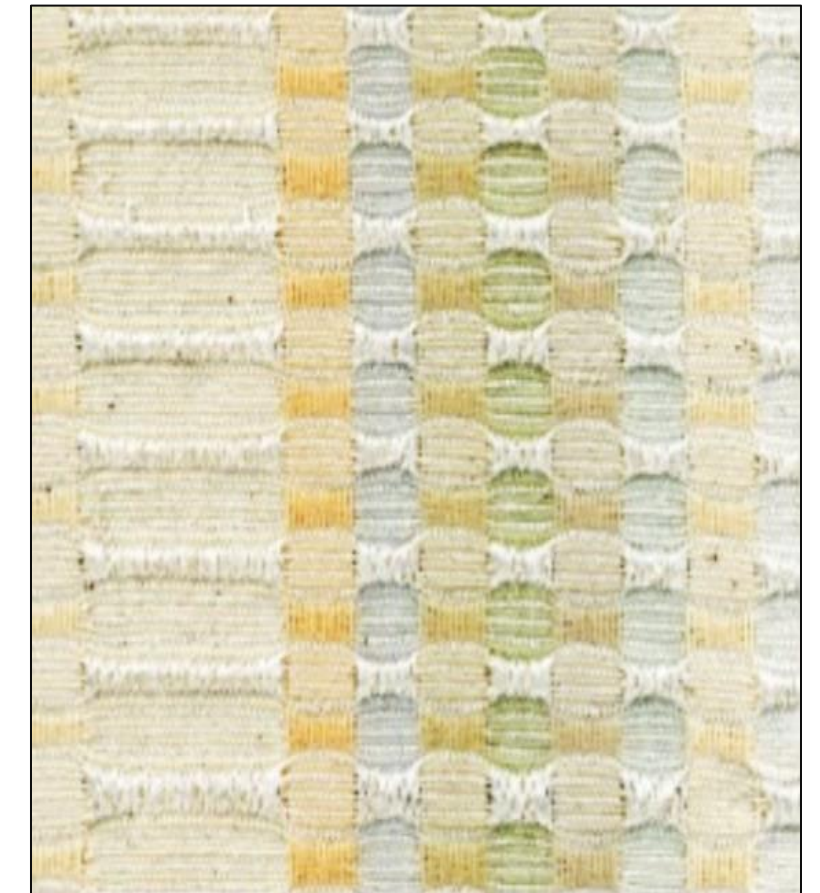
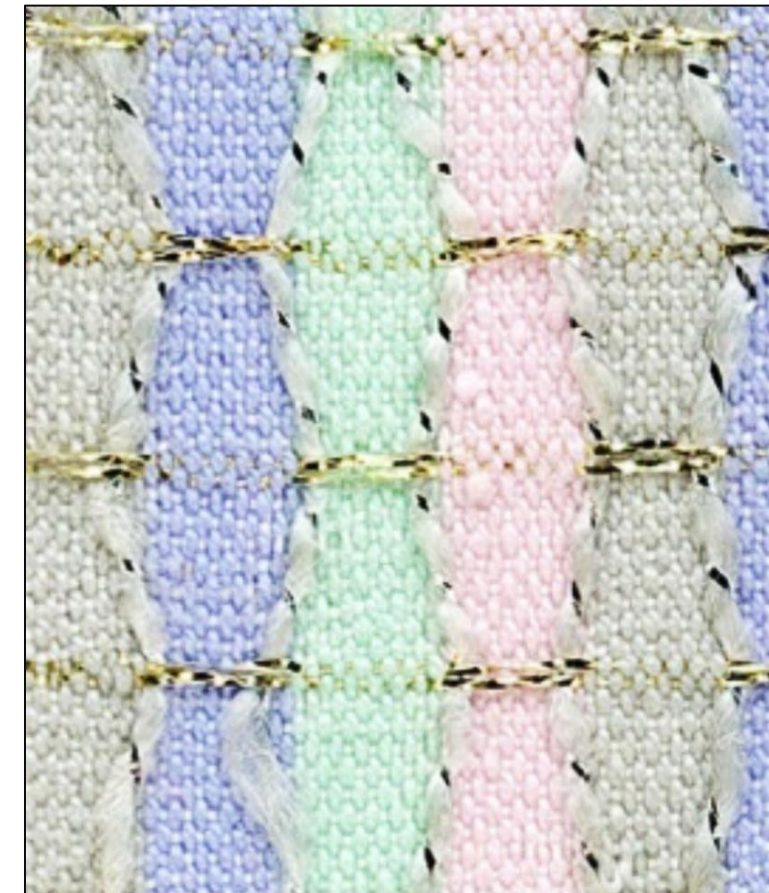
DISTORTED THREAD EFFECT

Features

1. Leno weave can also be used to create a figure by causing a *coarse ends to cross a number of standing ends*.
2. Crossing end is made to *lie at an angle to the normal direction* of the warp and weft threads.
3. *Latter must float freely* on the surface of the fabric.

Uses

1. Ornamental dress-wear, shirt.
2. Soft furnishing, Suits, Coats.



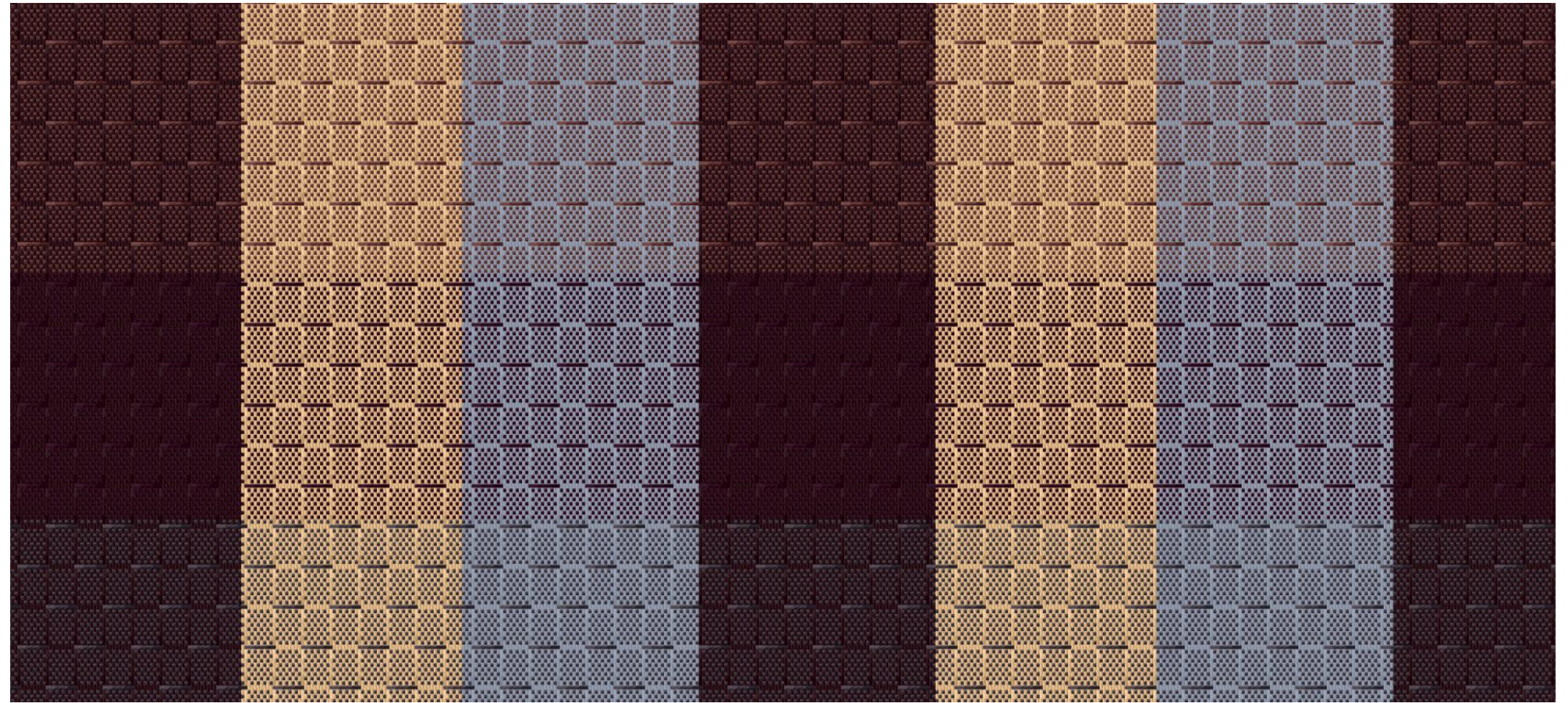
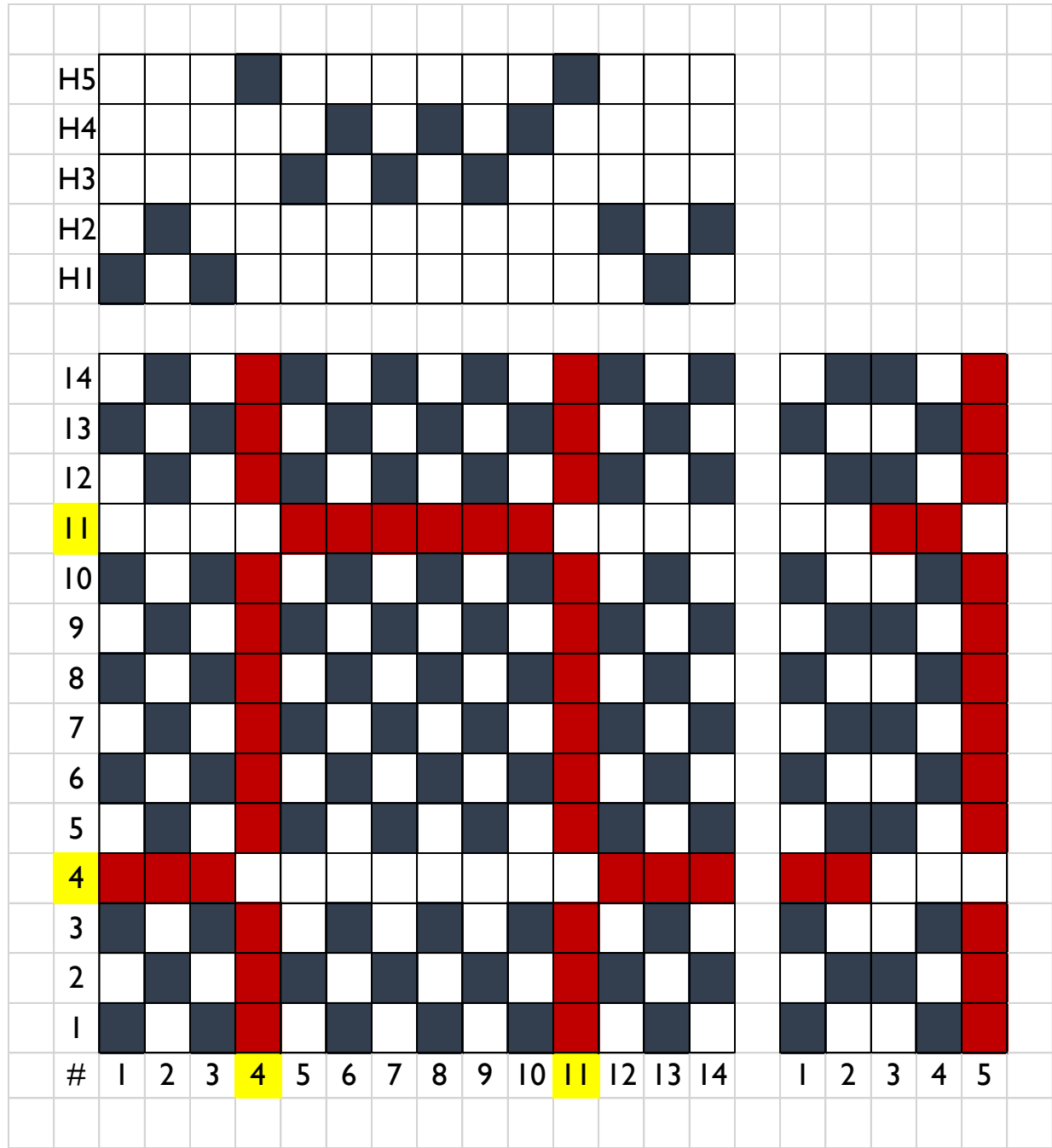
CONSTRUCTION PRINCIPLES

Stages in constructing the Distorted Warp weave:

1. After indicating the threads in the warp and weft direction which are essential for forming the distortion, fill in plain weave on all the remaining ends and picks, as at first stage.
2. For a warp distortion, lift the preselected warp threads of first stage except where they cross the preselected weft threads, and then lift all remaining ground ends over the preselected weft threads in one group on the first pick and in the other group on the second pick, as at second stage.
3. The completed design, third stage, is then formed by combining first and second stage.

14														
13														
12														
11														
10														
9														
8														
7														
6														
5														
4														
3														
2														
1														
#	1	2	3	4	5	6	7	8	9	10	11	12	13	14

DISTORTED THREAD EFFECT

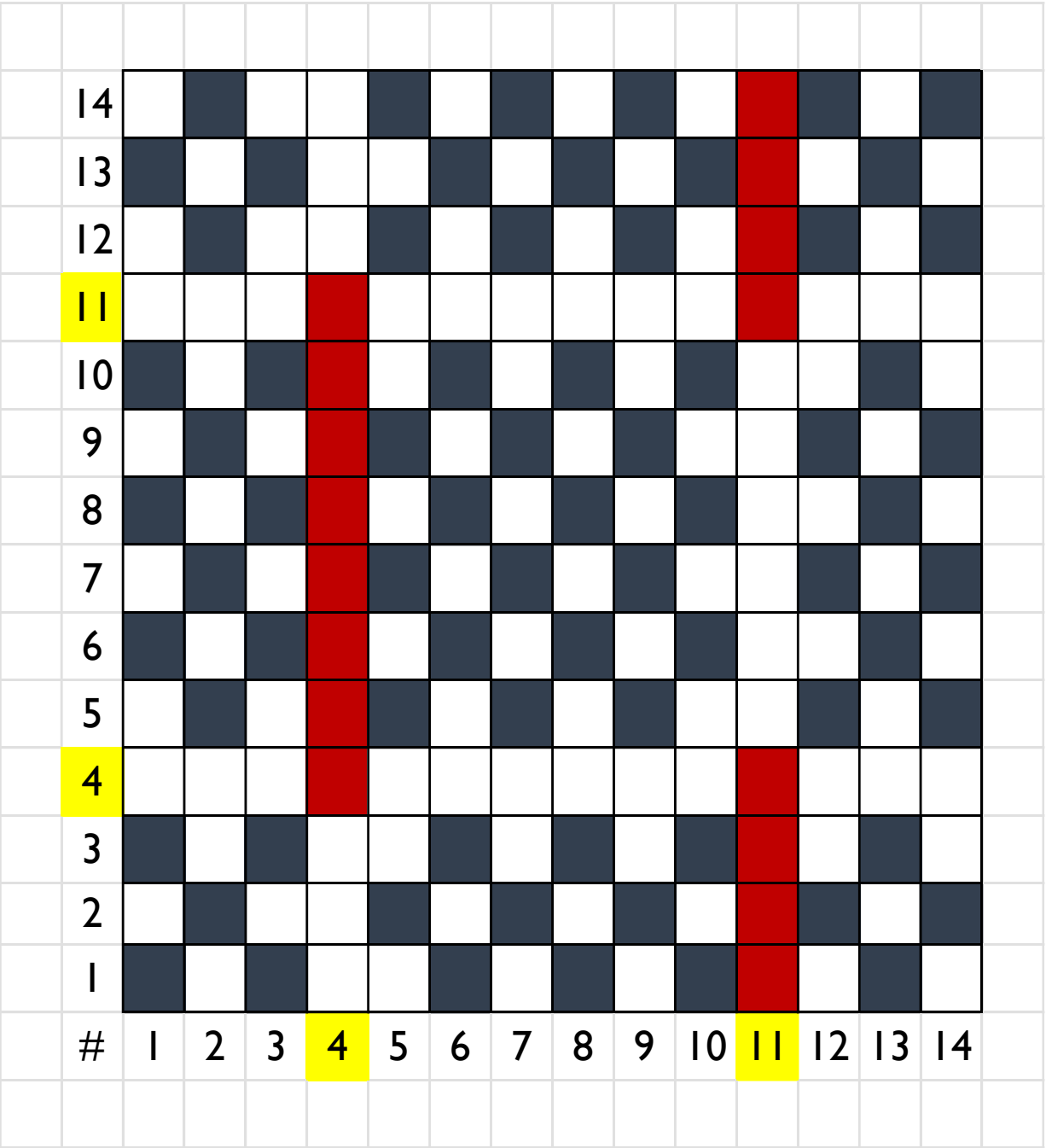


DISTORTED WARP

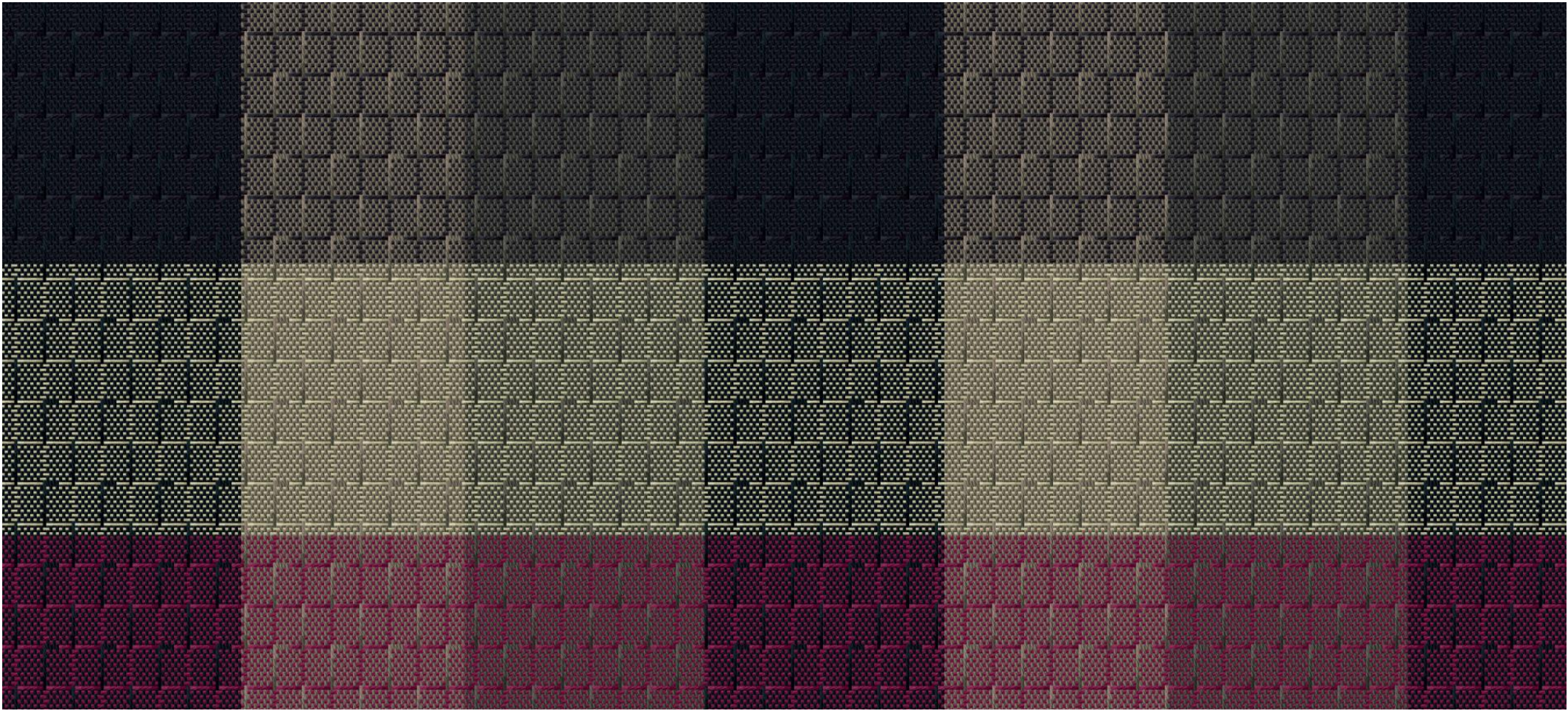
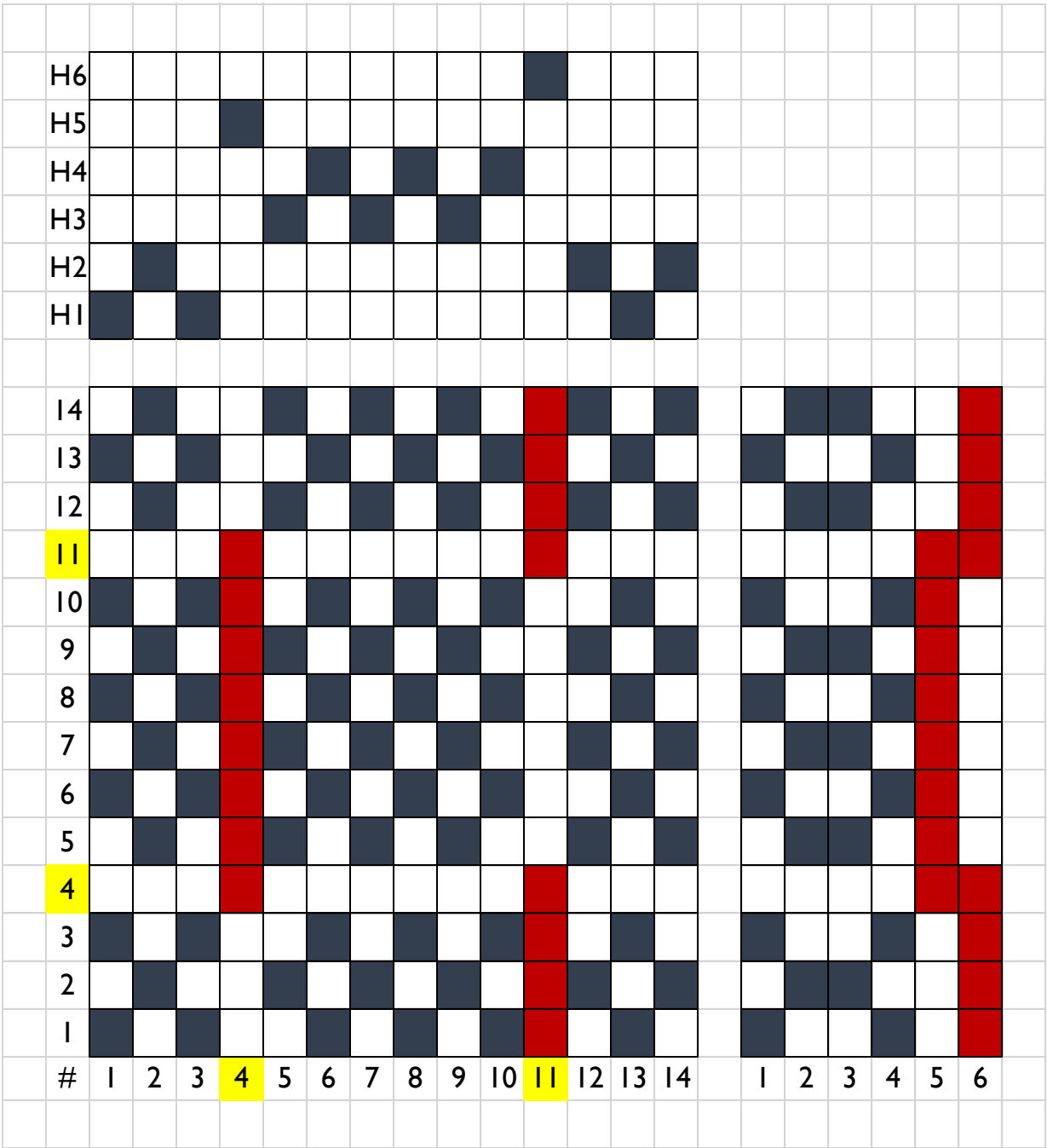
CONSTRUCTION PRINCIPLES

Stages in constructing the Distorted Weft weave:

- 1. After indicating the threads in the warp and weft direction which are essential for forming the distortion, fill in plain weave on all the remaining ends and picks, as at first stage.
- 2. For a weft distortion, lift the preselected weft threads of first stage except where they cross the preselected warp threads, and then lift all remaining ground picks over the preselected warp threads in one group on the first end and in the other group on the second end, as at second stage.
- 3. The completed design, third stage, is then formed by combining first and second stage.



DISTORTED THREAD EFFECT



DISTORTED WEFT

BEDFORD CORD

Features

1. Produce *longitudinal warp line* in the cloth, with *sunken line between the cord*.
2. *Wadding or padding* used to give the *prominence* of cord effect.
3. *Ends and picks* are always even number.
4. *Picks number* is always 4. (i.e. 12x4, 16x4, 20x4)

Types

1. Plain face Bedford cord
2. Wadded Bedford cord
3. Crepon Bedford cord
4. Arranged with alternate picks
5. Twill face Bedford cord

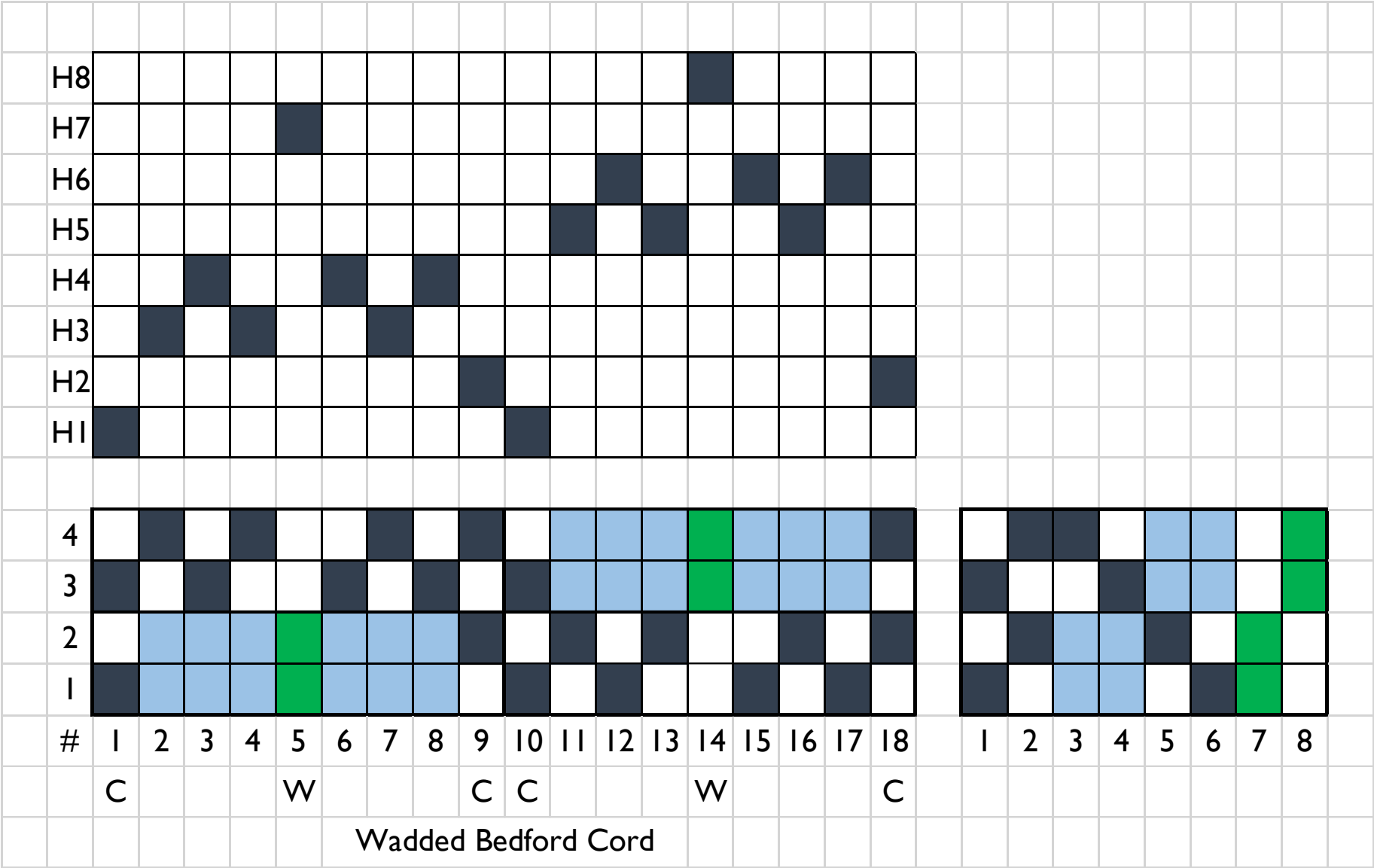
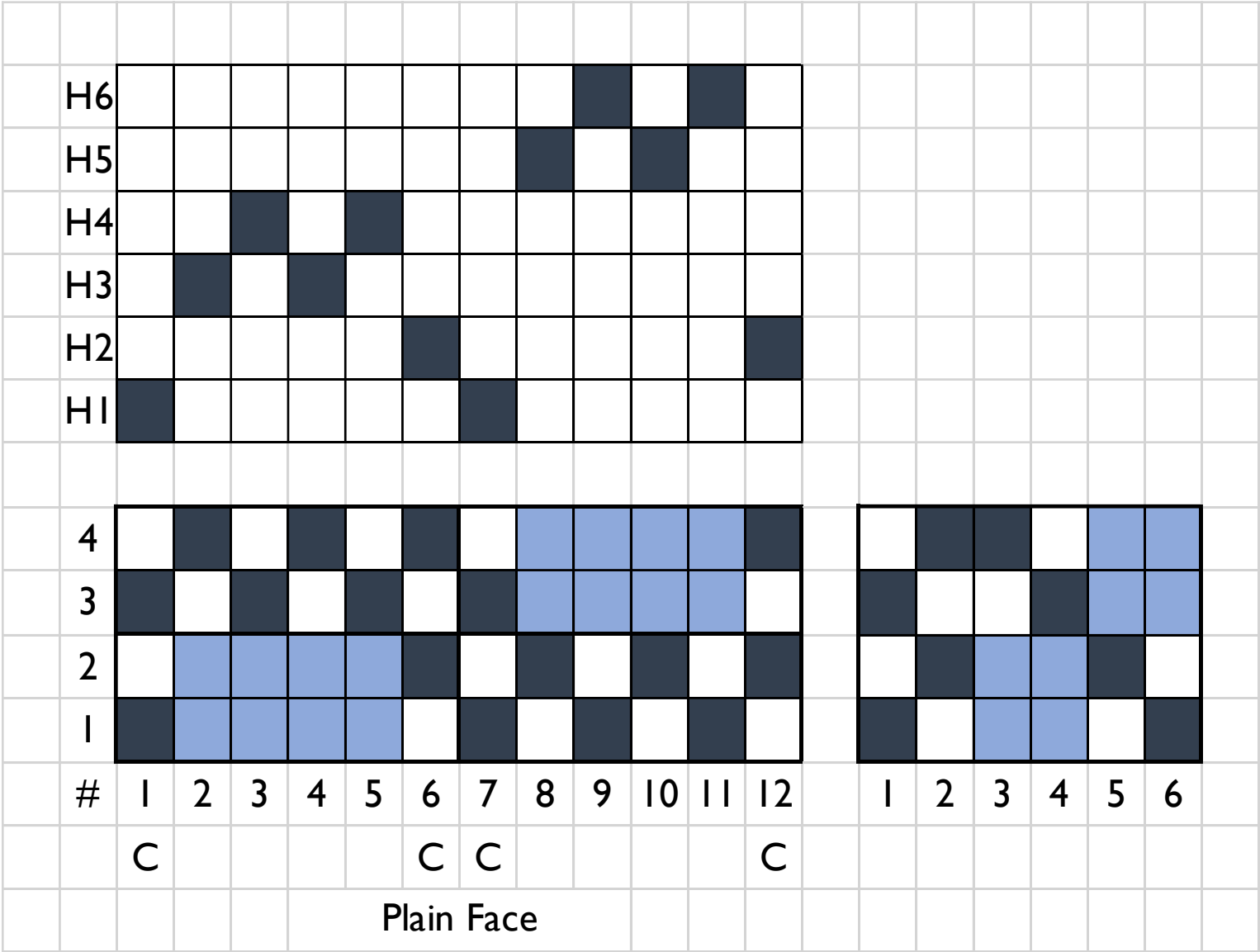
CONSTRUCTION PRINCIPLES

Stages in constructing the Plain Face Bedford Cord weave:

- 1. Indicate the width of two cords (in this example, each cord has six ends), and then show the outside ends of each cord, known as cutting ends/corner of the cord ends, weaving plain throughout, as at first and second stage.
- 2. The first pair of picks float under the warp ends in the first cord and weave plain in the second cord. The second pair of picks weave plain in the first cord but float under the warp ends, and thus on the back of the cloth on the second cord. This fourth stage is the final design.

	4																	
	3																	
	2																	
	1																	
	#	C						C	C								C	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	

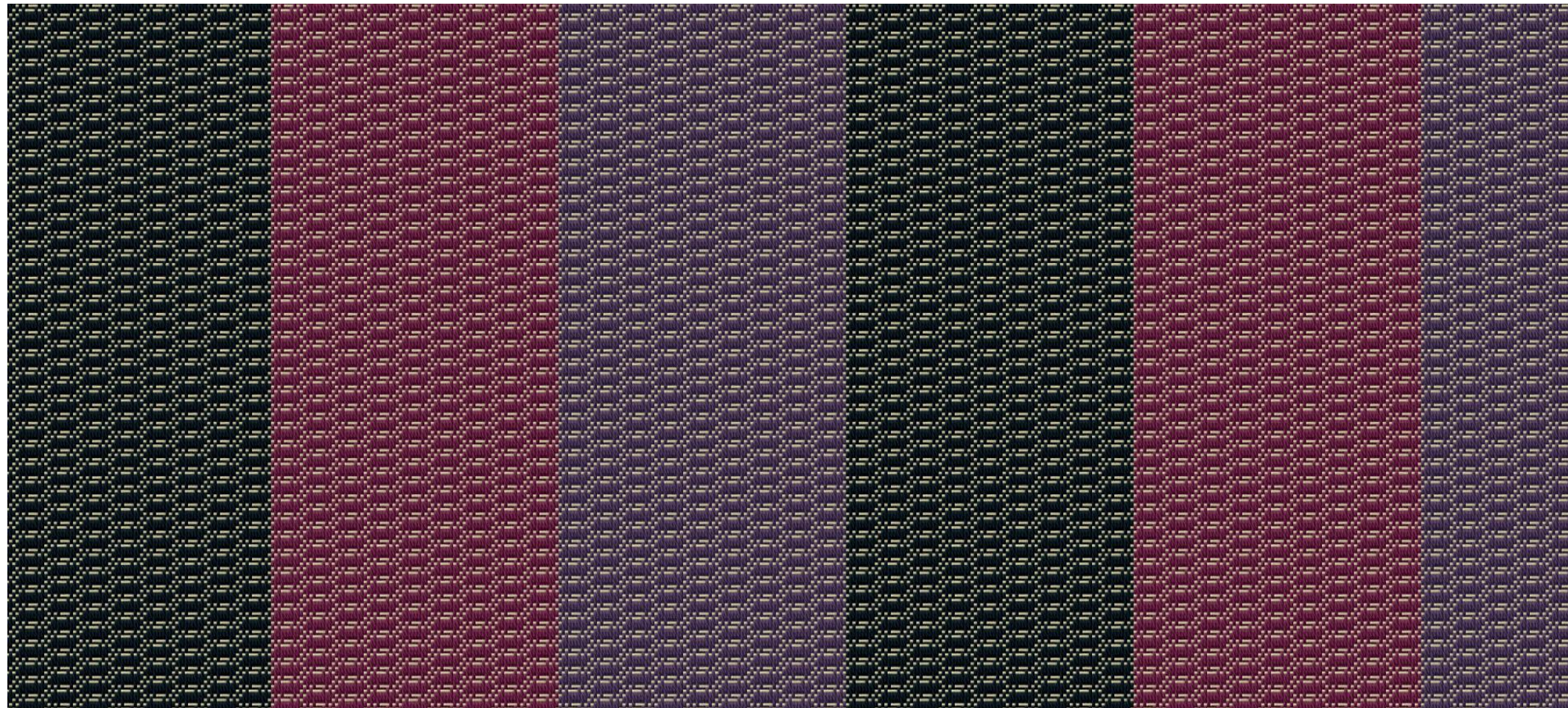
BEDFORD CORD



BEDFORD CORD

Uses

1. Dress-wear, Sportswear.
2. Soft furnishing, Ornamental trimmings.



PIQUE WEAVE

Features

1. A plain *face fabric composed of one series of warp and one series of weft thread.*
2. *Cords run in horizontally* in the cloth.
3. *One cord produce per repeat.*
4. *Skip denting system* is used to produce this weave.

Types

1. Ordinary Pique / Loose Back without wadding picks
2. Weft wadded welts / Loose Back wadded welt
3. Fast back welt
4. Waved pique

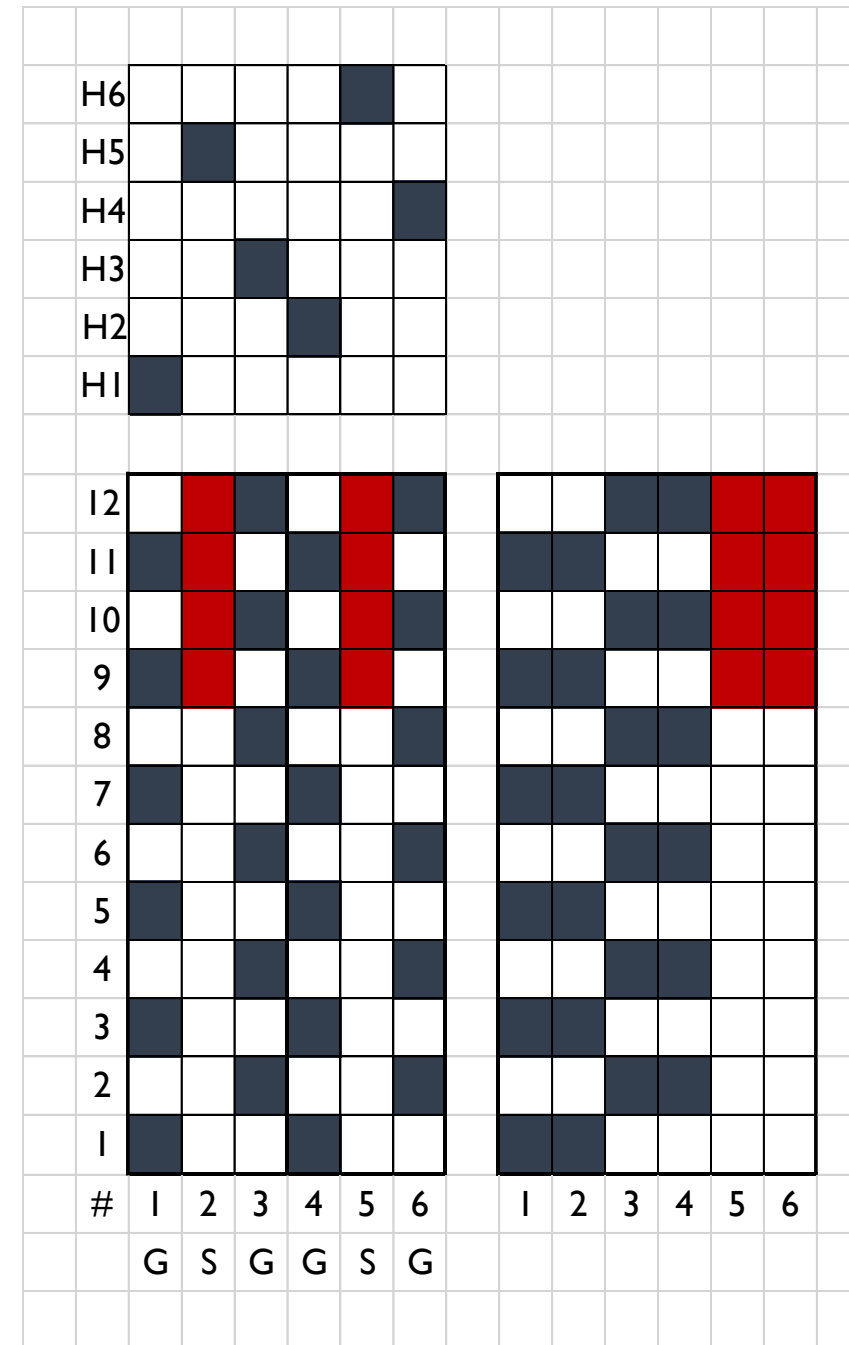
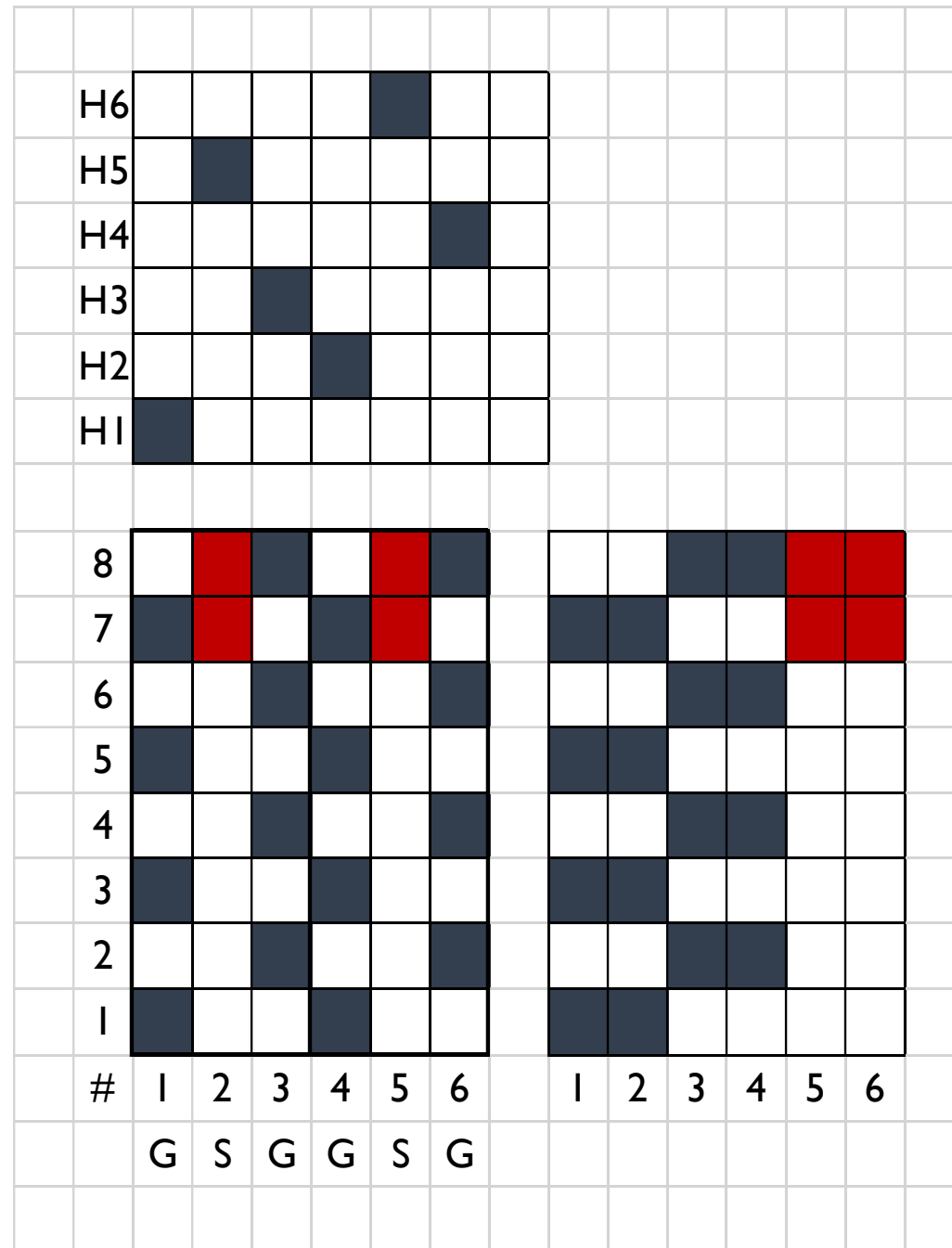
CONSTRUCTION PRINCIPLES

Stages in constructing the Ordinary Pique/Welt weave:

- 1. Indicate the order of the warp thread arrangement, which is always one ground, one stitching end and one ground end, then fill in plain weave on the ground ends as at first stage.
- 2. The stitching warp is lifted over the required number of picks, as determined by the requirements of the final fabric appearance; at second stage a two picks weave is illustrated.
- 3. The final weave is produced by combining first and second stage.

	8							
	7							
	6							
	5							
	4							
	3							
	2							
	1							
	#	1	2	3	4	5	6	
		G	S	G	G	S	G	

PIQUE WEAVE



ORDINARY PIQUE/LOOSE BACK WITHOUT WADDING PICKS

CONSTRUCTION PRINCIPLES

Stages in constructing the Wadded Welt weave:

1. In order to increase the prominence of the unstitched portions of the cloth, i.e. horizontal cords and to make the cloth more substantial, it is customary to *insert wadding picks between the tight back stitching ends and the slack face fabric.*
2. Usually, the *wadding weft is thicker than the ground weft*, and is inserted two picks at a place, the looms being provided with *changing shuttle boxes at one side only*. Sometimes, however, the same kind of weft is used for both the face and the wadding, looms with a single box at each side being employed; and, in such a case, one wadding pick at a place may be inserted.

		8					
		7					
		6					
		5					
W	4						
W	3						
	2						
	1						
#		1	2	3	4	5	6
		G	S	G	G	S	G

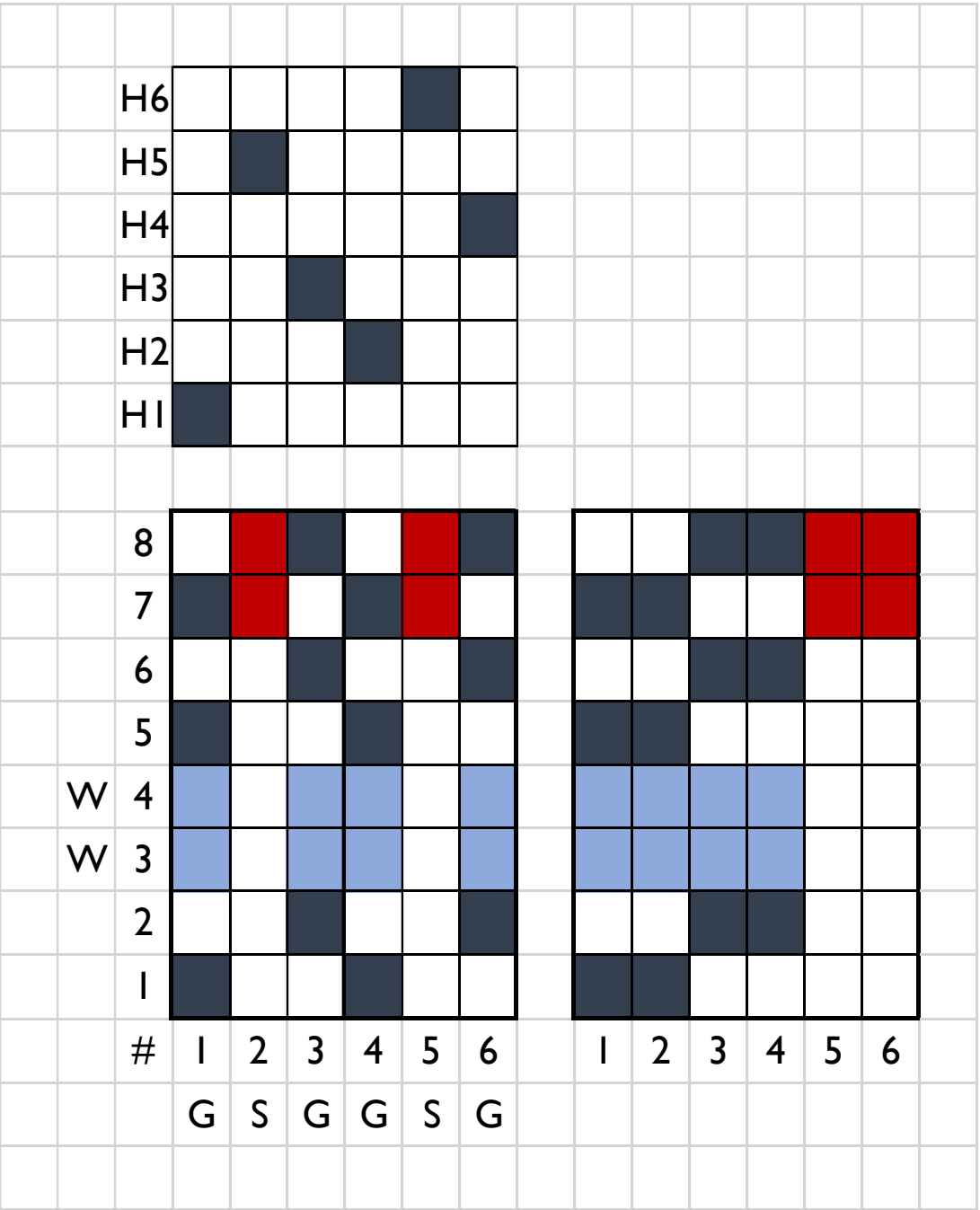
CONSTRUCTION PRINCIPLES

Stages in constructing the Fast Back Welt weave:

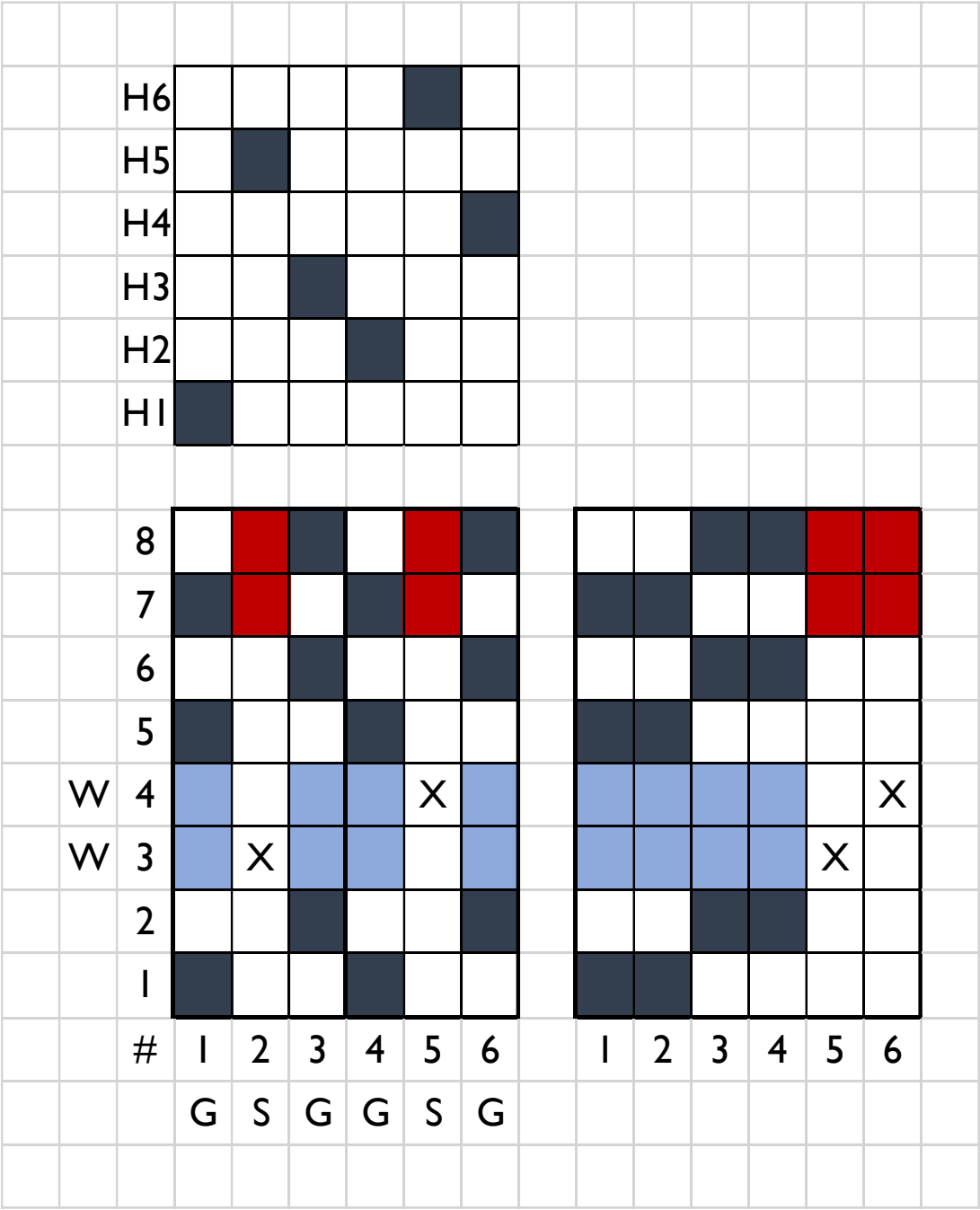
1. In each of the foregoing designs, the stitching ends are only lifted to form the indentations, the term "loose-back" being applied to this type of structure.
2. The term "fast-back" is applied to cloths in which the stitching ends are interwoven in plain order with all, or some wadding picks. The reduction of the float length of the stitching ends on the back of the fabric which results from this interlacing helps. to produce a more serviceable cloth less liable to accidental damage.

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PIQUE WEAVE



WEFT WADDED WELT/LOOSE BACK WADDED WELTS



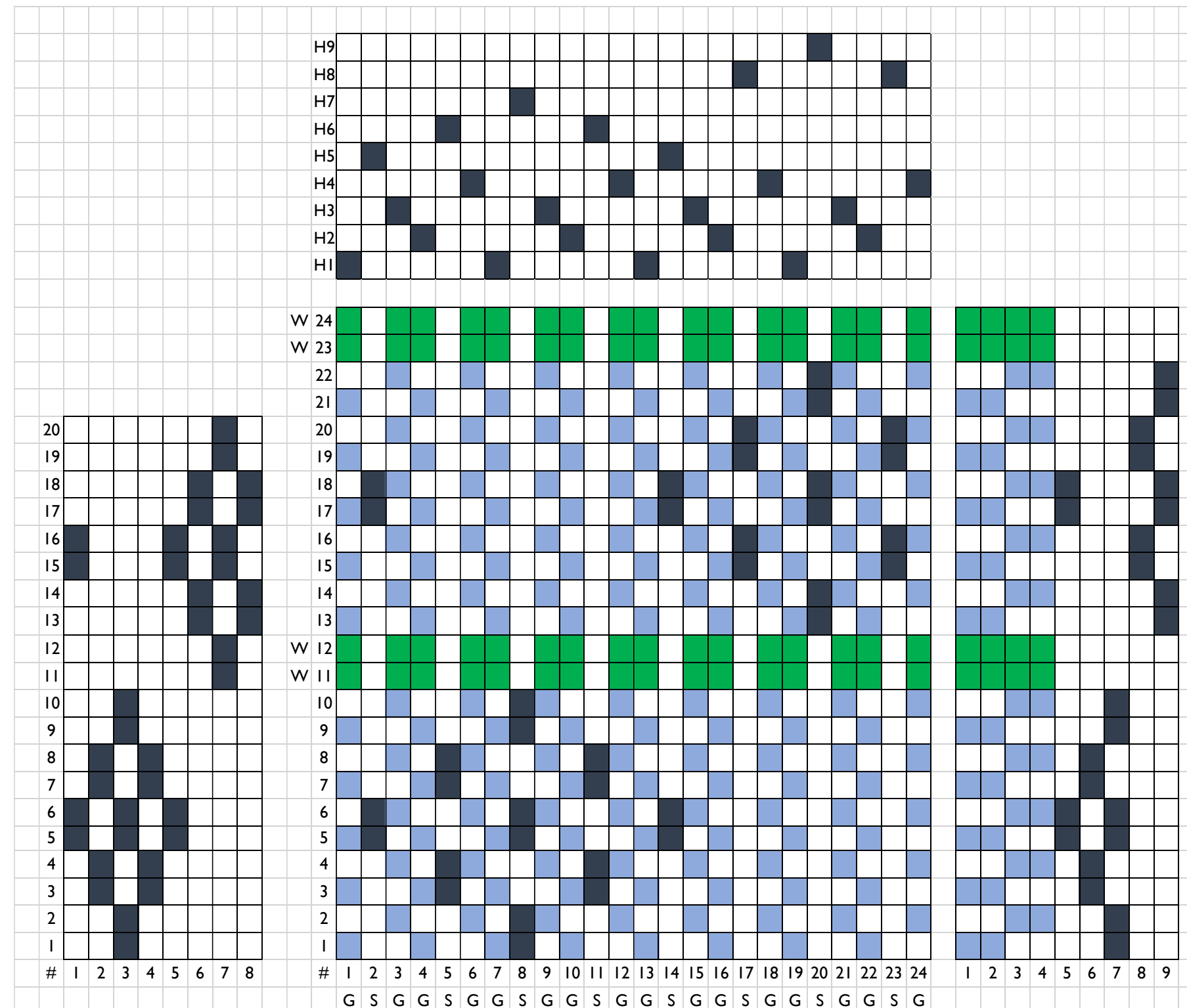
FAST BACK WELT

CONSTRUCTION PRINCIPLES

Stages in constructing the Waved Pique weave:

1. A waved pique is a simple modification of the welt structure in which the indentations are not in a horizontal line but are arranged in alternate groups, the marks in which indicate the lifts of the stitching ends on the face picks.
2. The group of marks does not overlap horizontally, as one commences on a face pick immediately following that on which the other has finished.
3. Between succeeding groups two wadding picks are inserted.
4. The complete design to correspond with first figure is given at second figure, in which the ends are arranged in the same order as in a welt, while there are ten face picks to two wadding picks. The lifts of the tight stitching ends force the wadding picks first in one direction and then in the other, so that waved lines are formed across the cloth. The following figures show the weave plan with drafting and lifting plan of a typical waved pique or welt design.

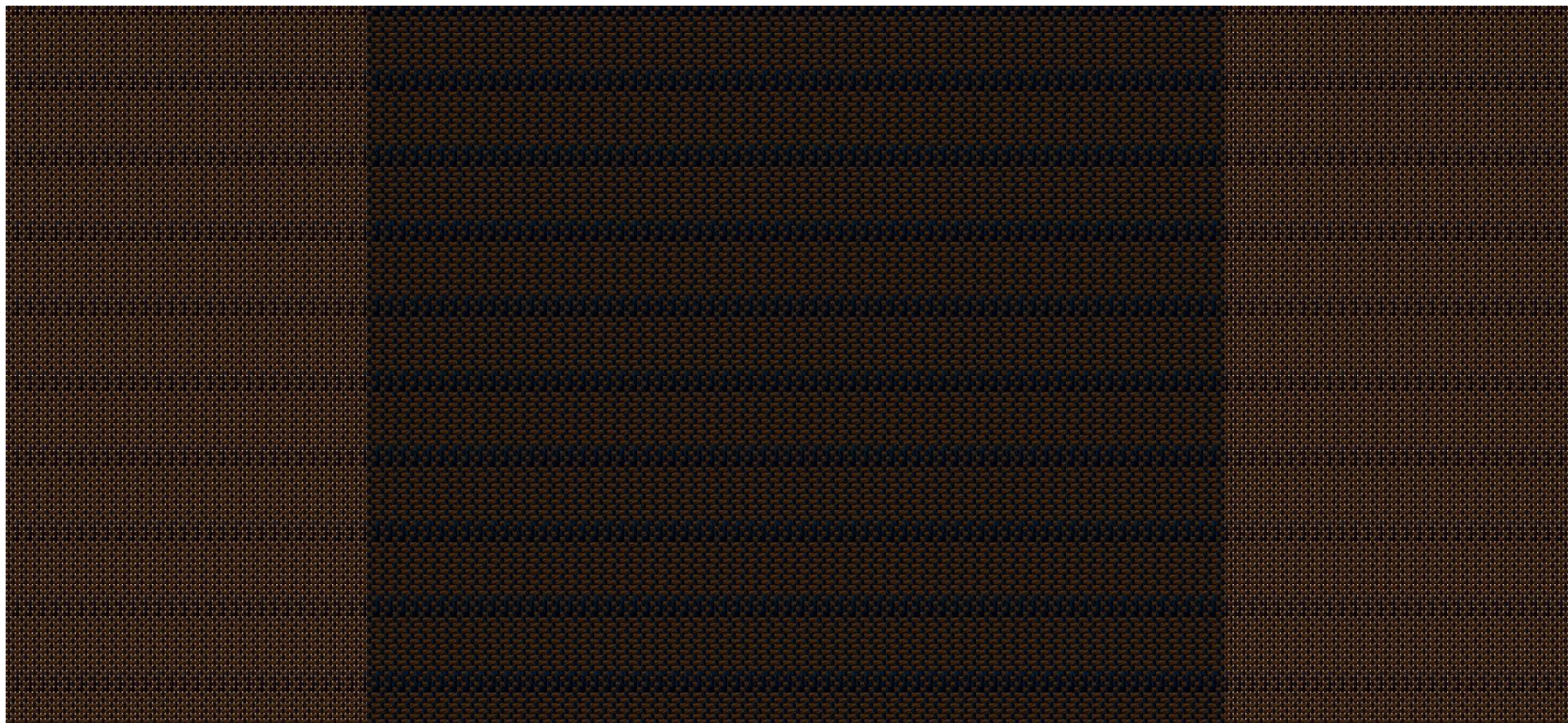
PIQUE WEAVE



PIQUE WEAVE

Uses

1. Trimmings, Ornaments
2. Neckties, Ladies light summer folding costume.



SPONGE WEAVE

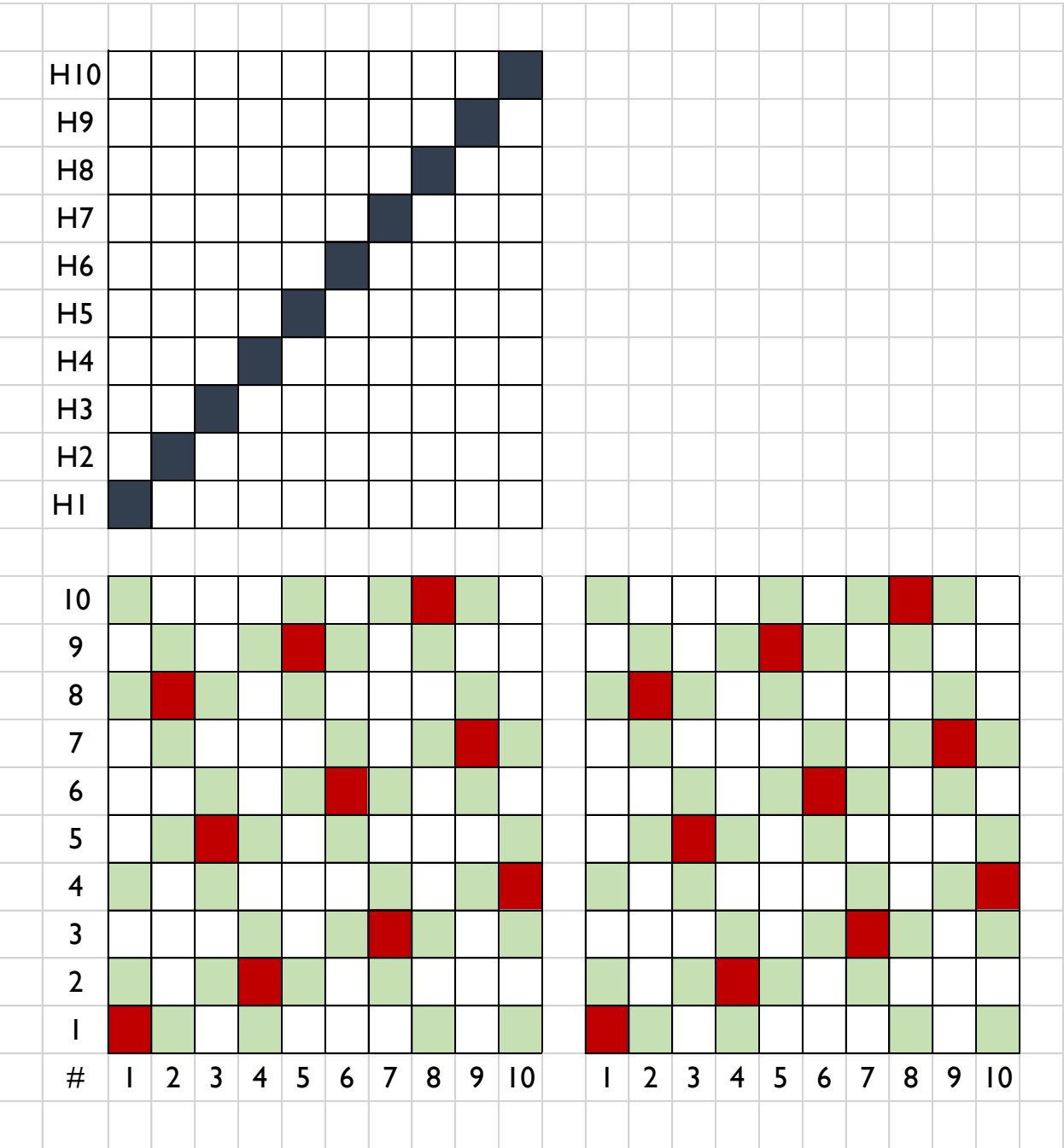
Features

1. Cellular structure and create a *soft spongy effect* in the fabric.
2. Number of *ends and picks are always equal*.
3. *10x10 is the smallest repeat size* of this weave.
4. *Straight drafting* system is used.
5. Low twisted and coarser yarns are used to produce this fabric.
6. Reversible cloth like honeycomb.
7. *Longest float of diamond = $\sqrt{\text{Number of ends or picks in a repeat} - 1}$*
8. This weave is produced on the *sateen based*.
9. Also used as *ground of jacquard design*.

SPONGE WEAVE

Application

- 1. Woolen shawls.
- 2. Bed-sheet.
- 3. Towel
- 4. Drapes
- 5. Dress fabric
- 6. Bathing wraps



CREPE WEAVE

Four basic methods of producing crepe weaves:

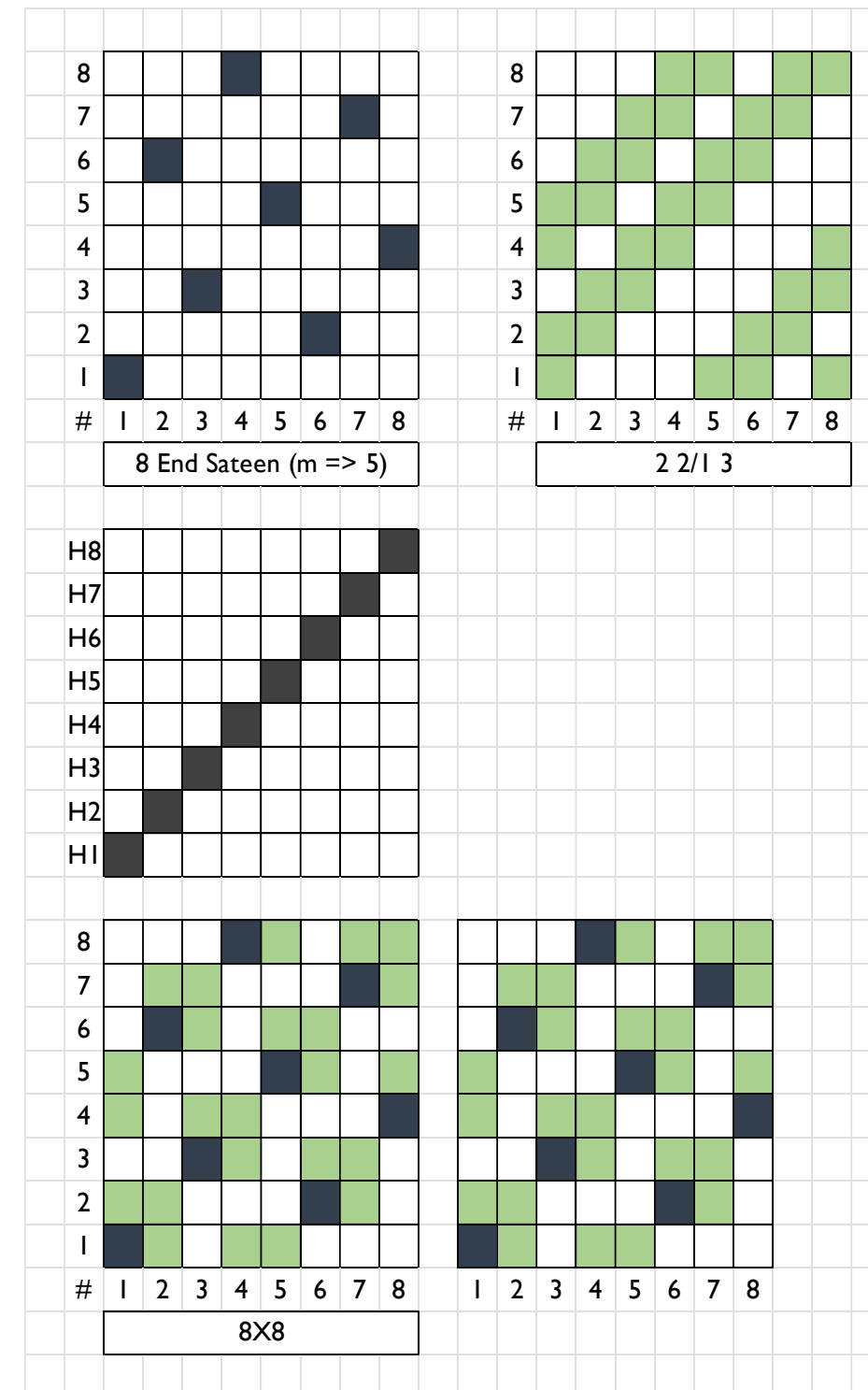
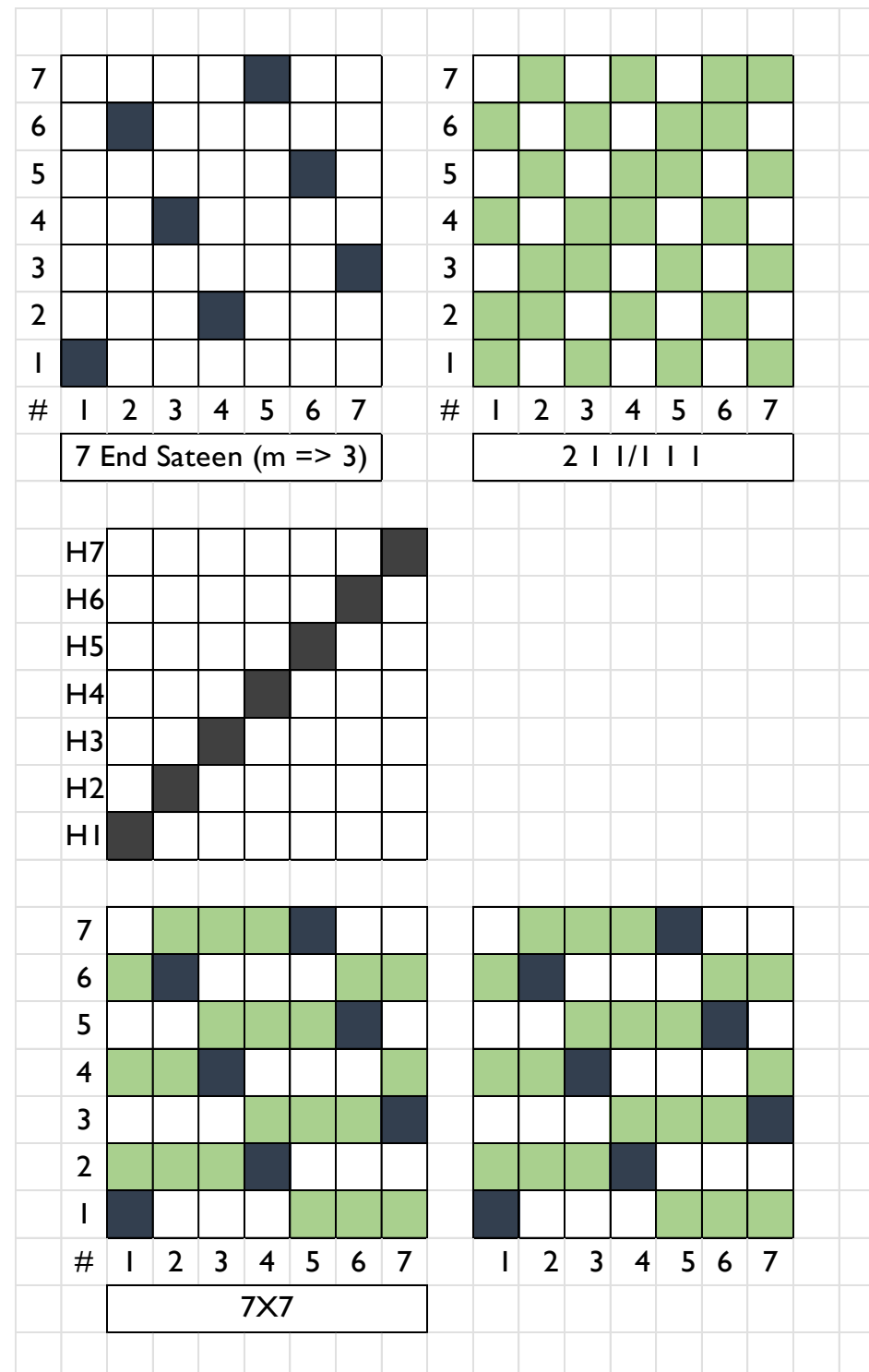
1. On a Sateen Base
2. By Reversing
3. By Superimposing
4. On a Plain weave base

CREPE WEAVE – SATEEN BASE

Methods of Construction:

1. Construct a sateen weave.
2. Construct a twill weave on the same repeat size.
3. Using the sateen base as the starting point of each lift of twill, rearrange the twill weave on the sateen base.
4. This new weave is called sateen base crepe weave.

CREPE WEAVE – SATEEN BASE

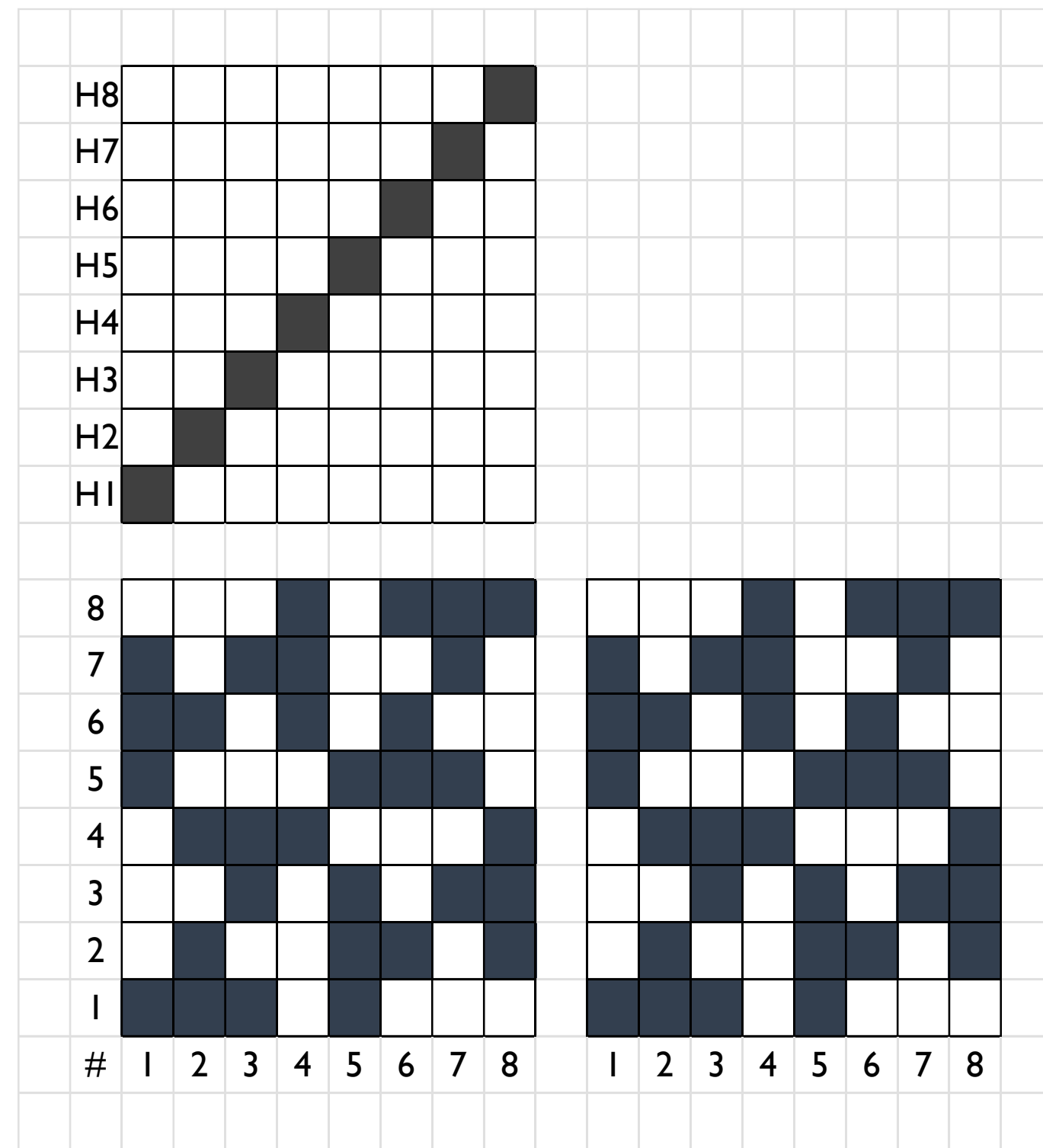
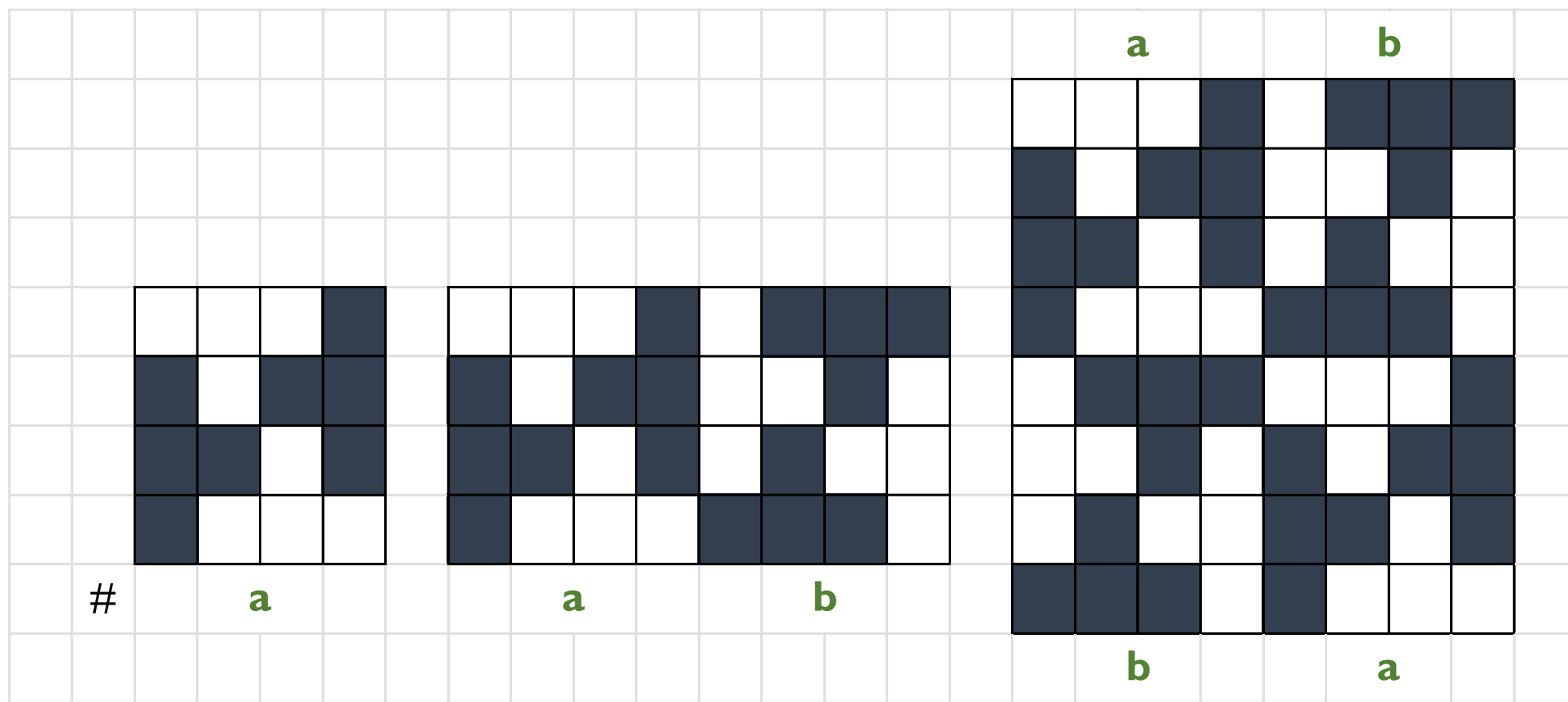


CREPE WEAVE – REVERSING

Methods of Construction:

1. Make a small motif, as at 'a'.
2. Reverse 'a' by turning it over, so that the warp lifts of the 4th end become the weft lifts of the 5th end and those of the 4th end become the warp lifts of the 5th end; similarly, the 6th, 7th, and 8th ends are the converse of the 3rd, 2nd and 1st respectively, and the design is now on 8 ends x 4 picks.
3. Reverse this by turning it over in the weft direction and using the same technique as described. The final design is thus produced, which is called crepe weave.

CREPE WEAVE – REVERSING



CREPE WEAVE – SUPERIMPOSING

Methods of Construction:

1. Construct two different weaves with same repeat size.
2. Superimpose one weave on the other.

CREPE WEAVE – SUPERIMPOSING

The figure displays two 5x5 grids, each representing a different weave pattern. The rows are labeled 1 to 5 on the left, and the columns are labeled # 1 to 5 at the bottom. The first grid, labeled '5 End Sateen', shows a dark blue pattern where the main diagonal cells (1,1), (2,2), (3,3), (4,4), and (5,5) are shaded. The second grid, labeled '1 1/1 2', shows a light green pattern where the cells at (1,1), (1,4), (2,2), (2,5), (3,3), (4,2), (4,5), and (5,1) are shaded.

	# 1	2	3	4	5
5				■	
4		■			
3					■
2			■		
1	■				

5 End Sateen

	# 1	2	3	4	5
5			■		■
4		■		■	
3	■		■		
2		■			■
1	■			■	

1 1/1 2

The figure displays two 5x5 matrices, each representing a system's state over time. The left matrix is labeled with rows H1 to H5 and columns 1 to 5. The right matrix is labeled with rows 1 to 5 and columns 1 to 5. Both matrices show a diagonal pattern of dark gray cells, indicating a system where the state at time t depends only on the state at time t-1.

Left Matrix (H1 to H5):

	1	2	3	4	5
H5					Dark Gray
H4				Dark Gray	
H3			Dark Gray		
H2		Dark Gray			
H1	Dark Gray				

Right Matrix (1 to 5):

	1	2	3	4	5
5			Light Green	Dark Gray	Light Green
4		Dark Gray		Light Green	
3	Light Green		Light Green		Dark Gray
2		Light Green	Dark Gray		Light Green
1	Dark Gray			Light Green	

Legend:

- Light Green: State at time t-1
- Dark Gray: State at time t

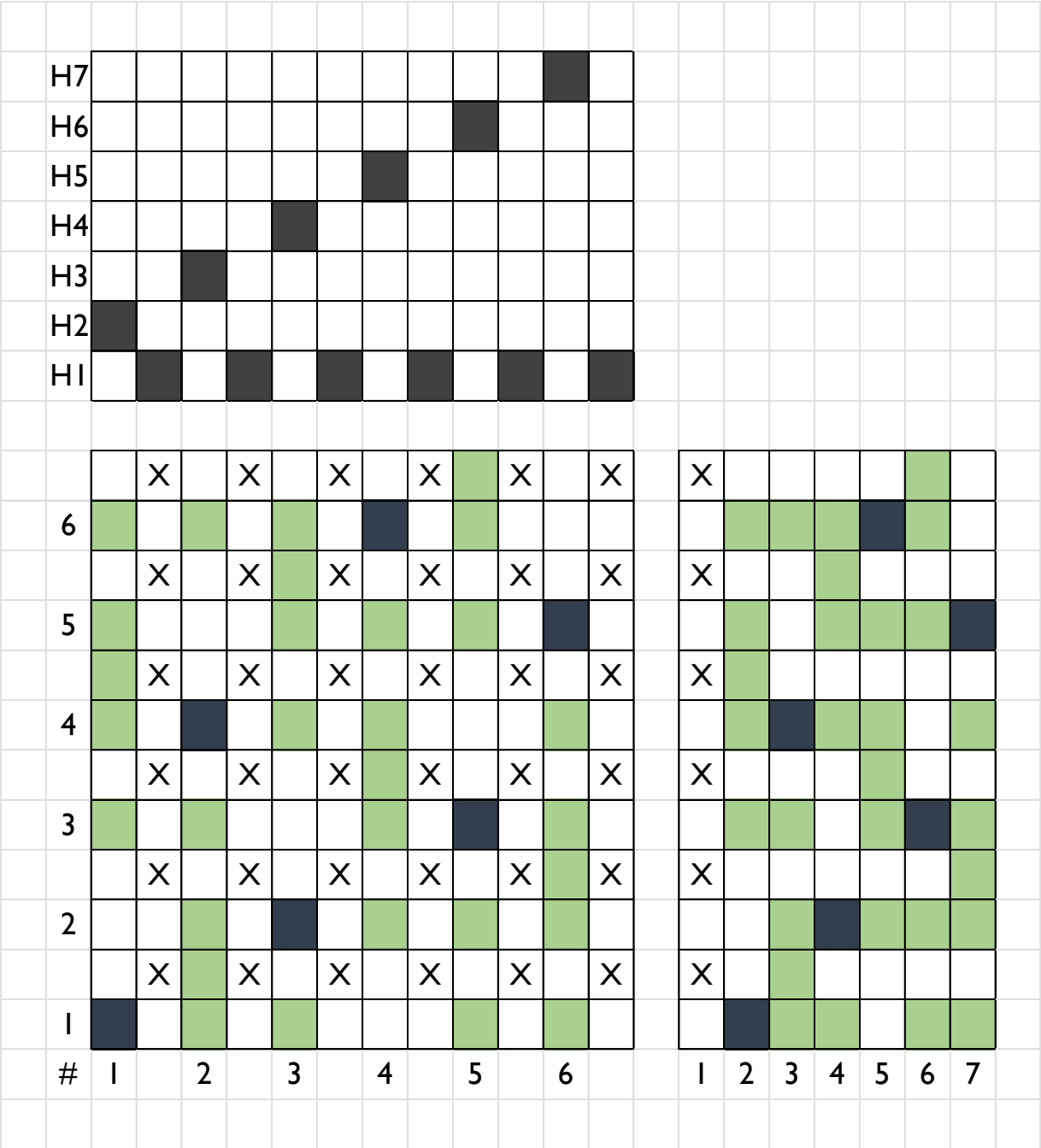
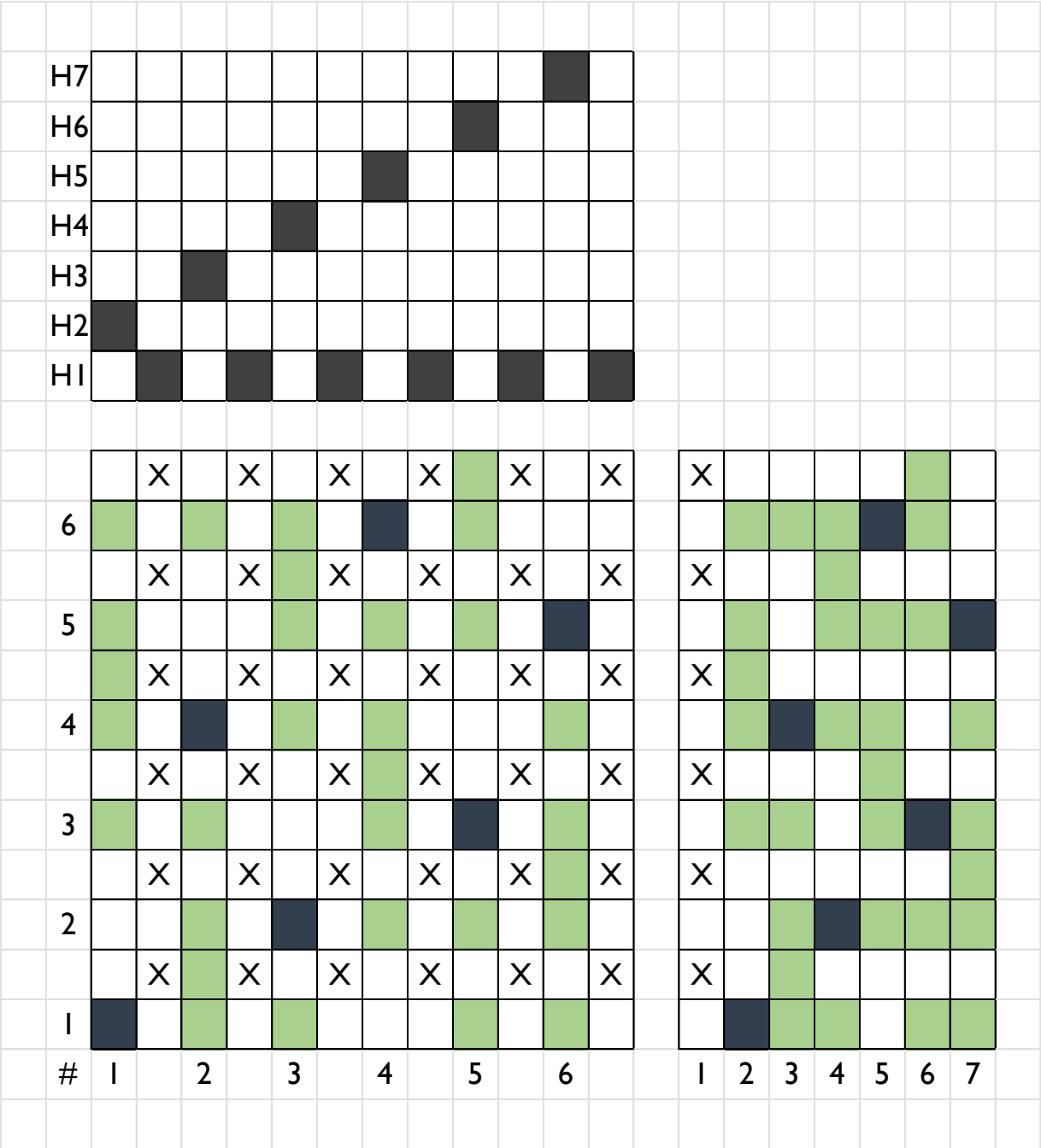
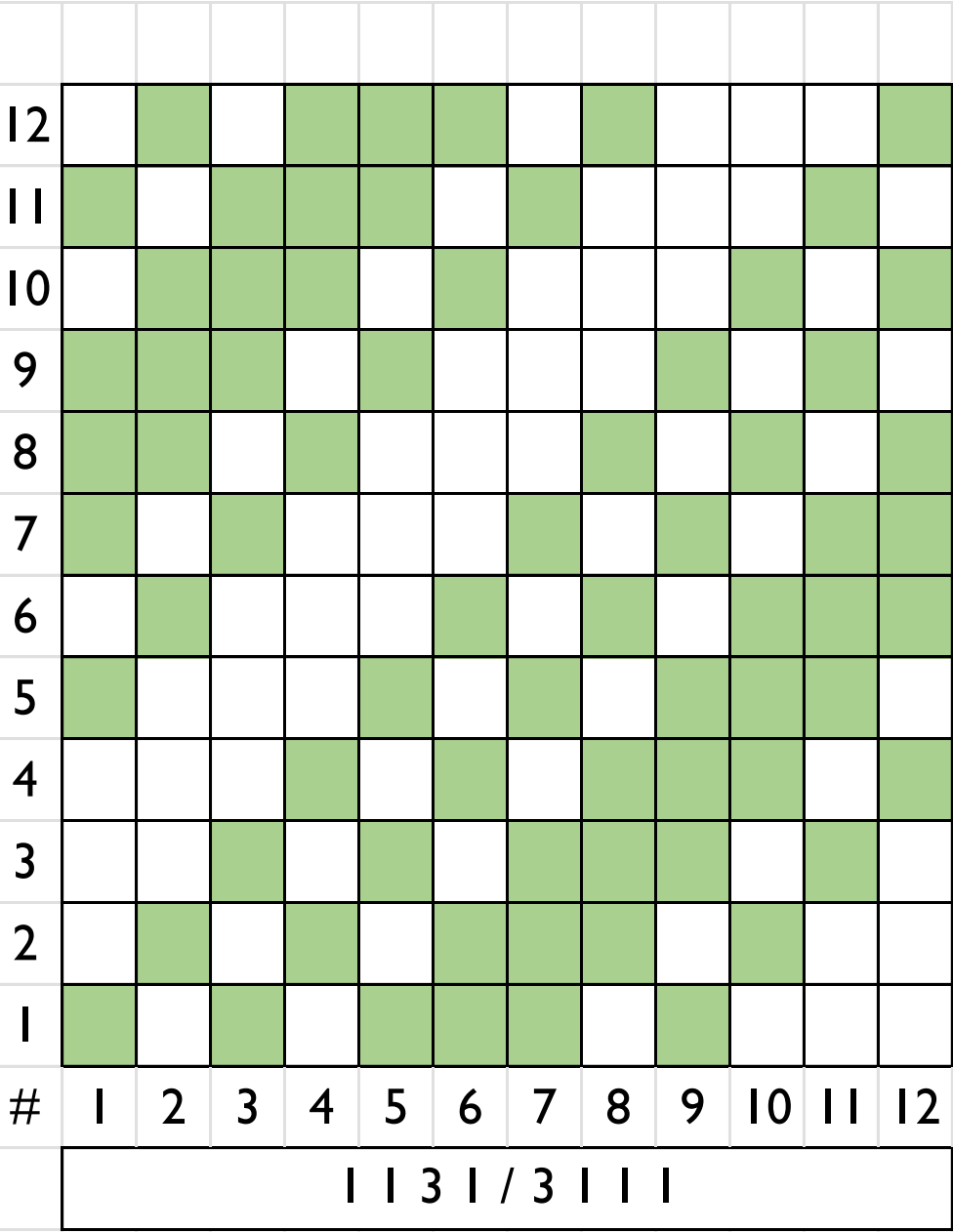
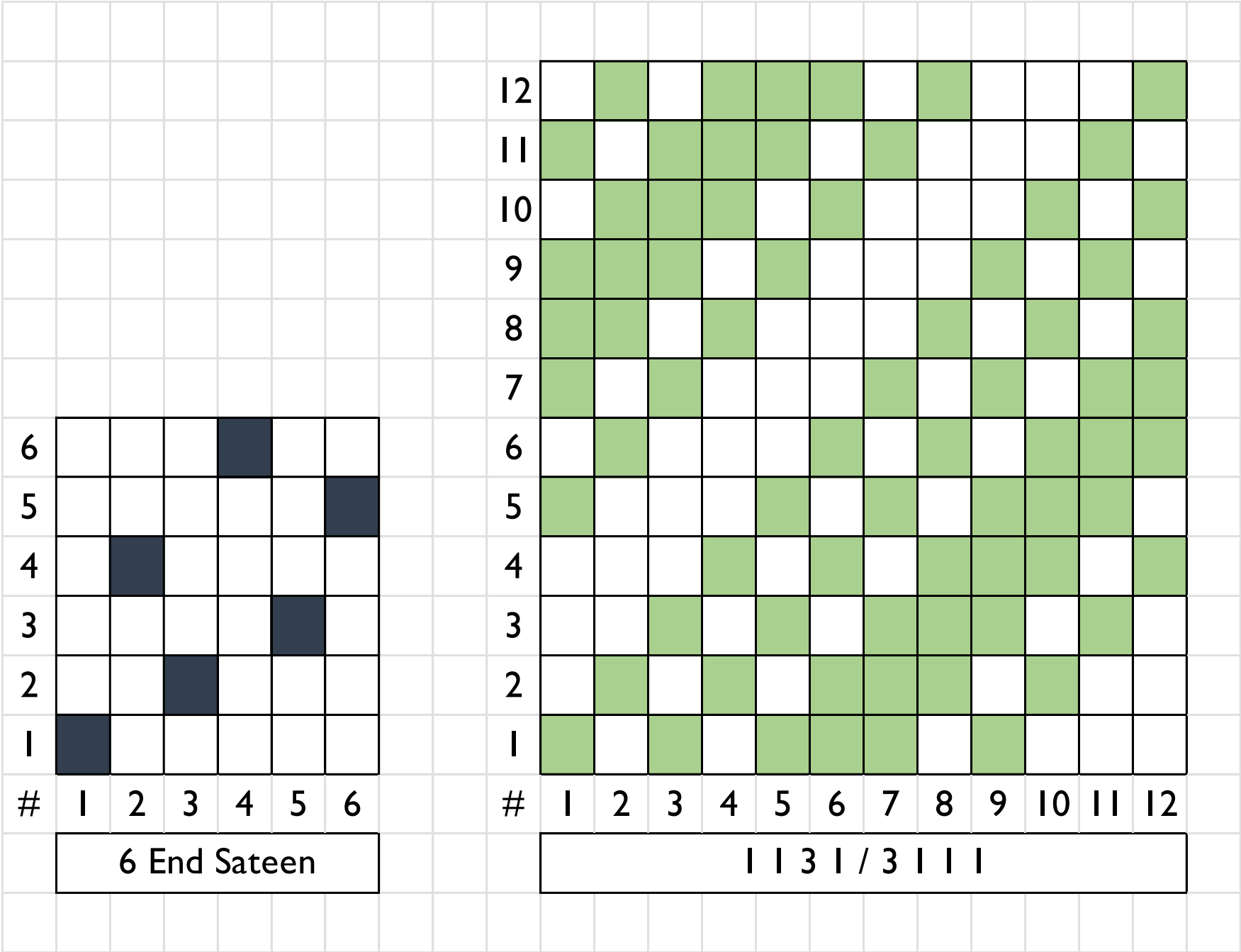
5X5

CREPE WEAVE – PLAIN BASE

Methods of Construction:

1. Design a sateen on half the number of ends and picks required in the final design a 6-end sateen will be used for a design to be produced on 12 ends x 12 picks.
2. Expand this weave so that the sateen base appears on alternate ends and picks only, and use this base as the starting point of each lift of a twill, in this case 1 1 3 1/3 1 1 1 as shown in the following figure.
3. On the remaining ends insert alternate ends of plain weave, i.e. all of these ends will weave the same tabby; care should be taken to lift the warp on the picks opposite to those on which the sateen base appears.
4. Now combine these to give the final design, which is repeated in the following figure.

CREPE WEAVE – PLAIN BASE



THANK
YOU !



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